

Cross-Wearer Gesture Recognition — LOUO Variant Comparison

Generated 2026-05-14 10:51:15

Data & configuration

LOUO path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
LOUO path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
LOUO path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
LOUO path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
LOUO path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
LOUO path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
LOUO path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
LOUO path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
LOUO path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
LOUO path 10	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
LOUO path 11	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Dynamic
LOUO path 12	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
LOUO path 13	/home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic
LOUO users loaded	13 (11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry, 8_Jestin)
Excluded classes	— (none)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Per-trial normalisation	none
Downstream normalisation	minmax
Sampling rate	30.0 Hz
Random state	42
Augmentation	off
Epochs · batch size	40 · 16
Early stopping	on · monitor val_loss · patience=8
LOUO inner-validation	user
LOUO variants	Baseline, Shallow, Deep, BN_GAP, WideKernel, CNN_LSTM, CNN_BiLSTM
Final-retrain variant	BN_GAP_Wide
Final-retrain held-out user	— (trained on all users)

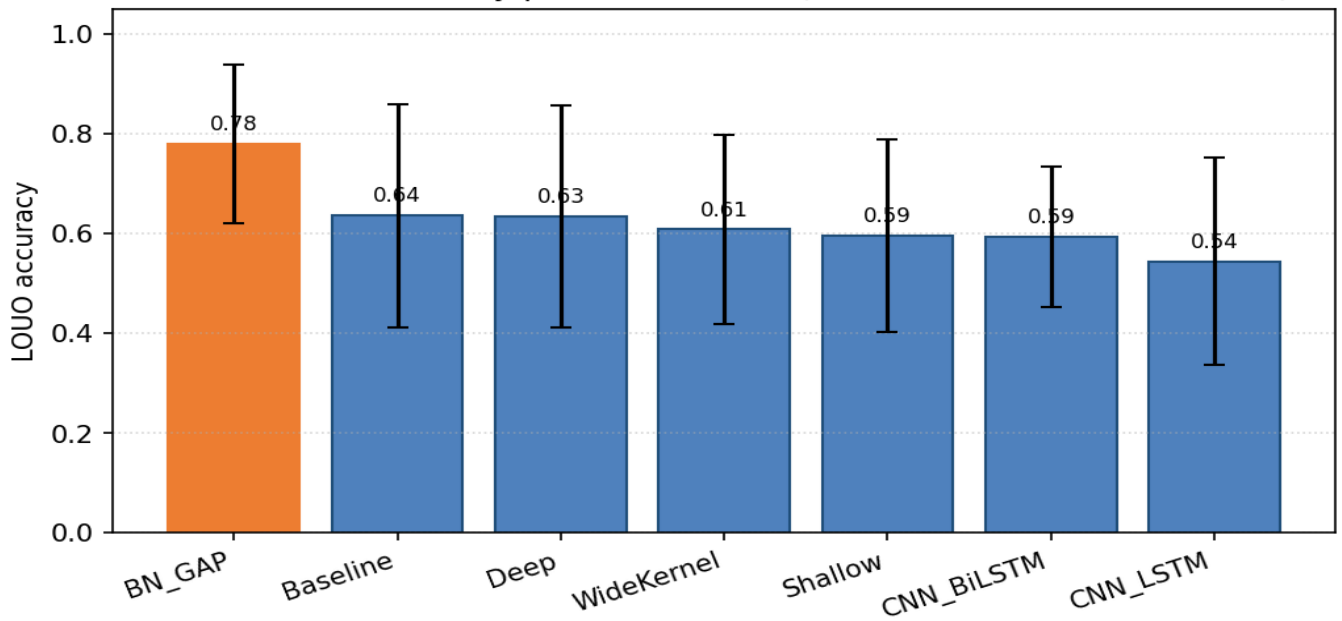
Variant comparison (LOUO)

Variant	Params	LOUO mean	LOUO std	Folds	Held-out user acc
BN_GAP	57,319	0.7790	0.1599	13	—
Baseline	113,255	0.6350	0.2243	13	—
Deep	204,007	0.6325	0.2225	13	—
WideKernel	131,687	0.6071	0.1891	13	—
Shallow	72,487	0.5945	0.1938	13	—
CNN_BiLSTM	105,575	0.5921	0.1405	13	—
CNN_LSTM	68,455	0.5431	0.2080	13	—

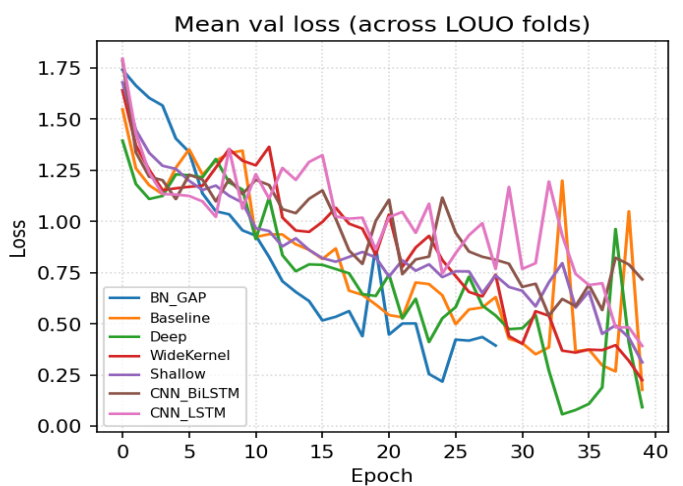
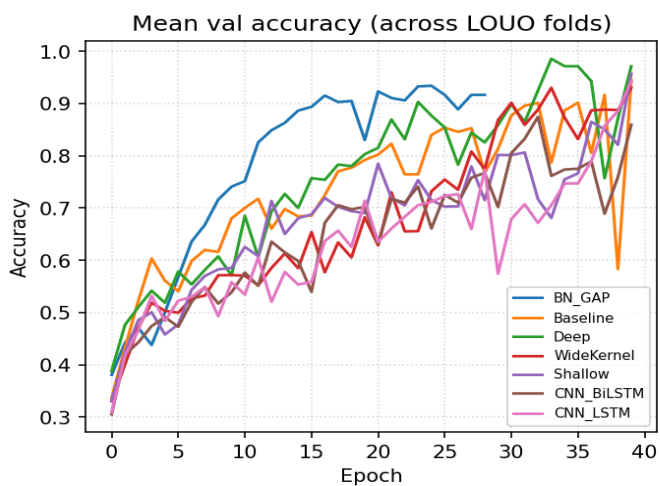
Best LOUO mean: BN_GAP

LOUO accuracy per variant

LOUO mean accuracy per architecture (error bars = std across folds)

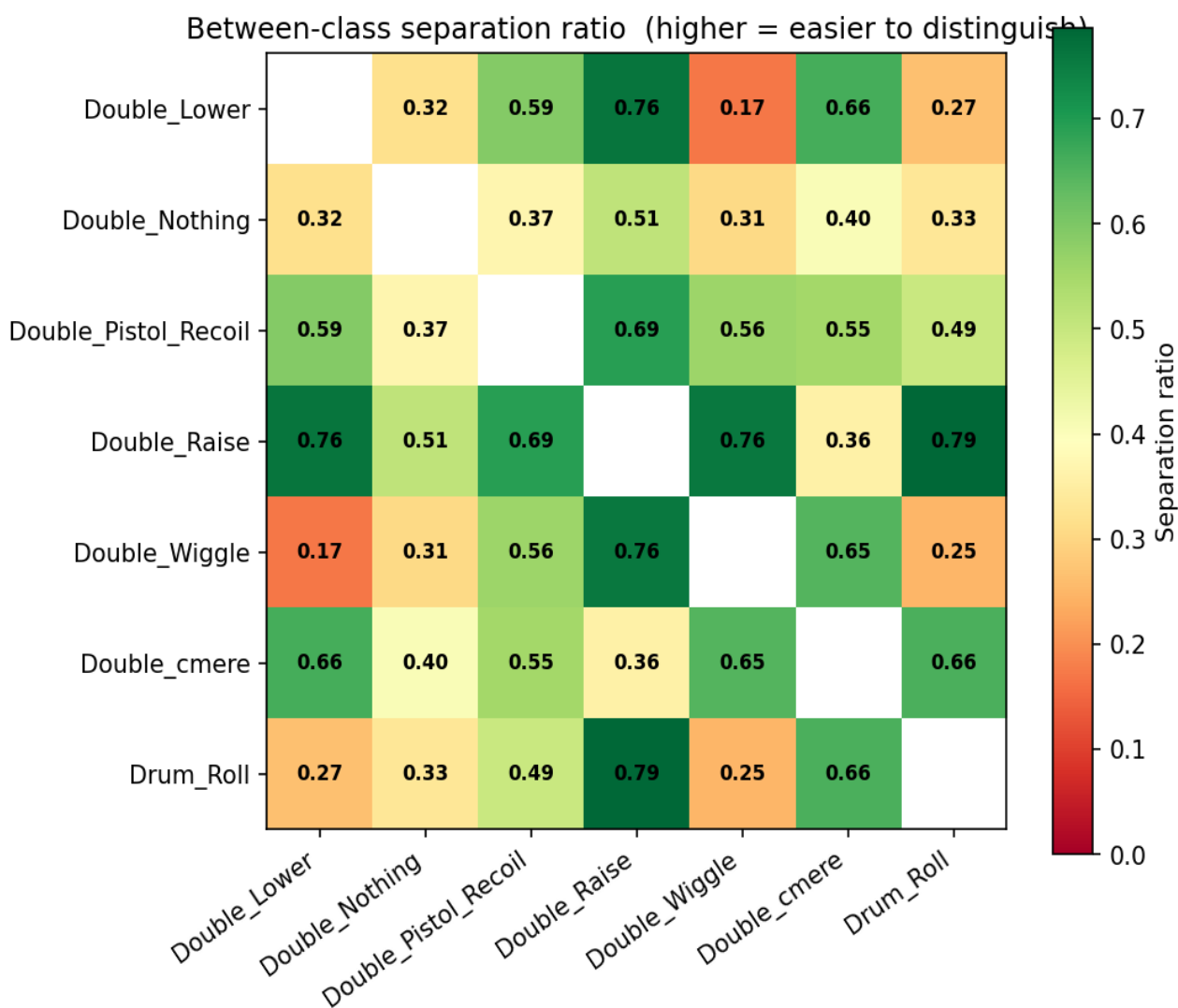
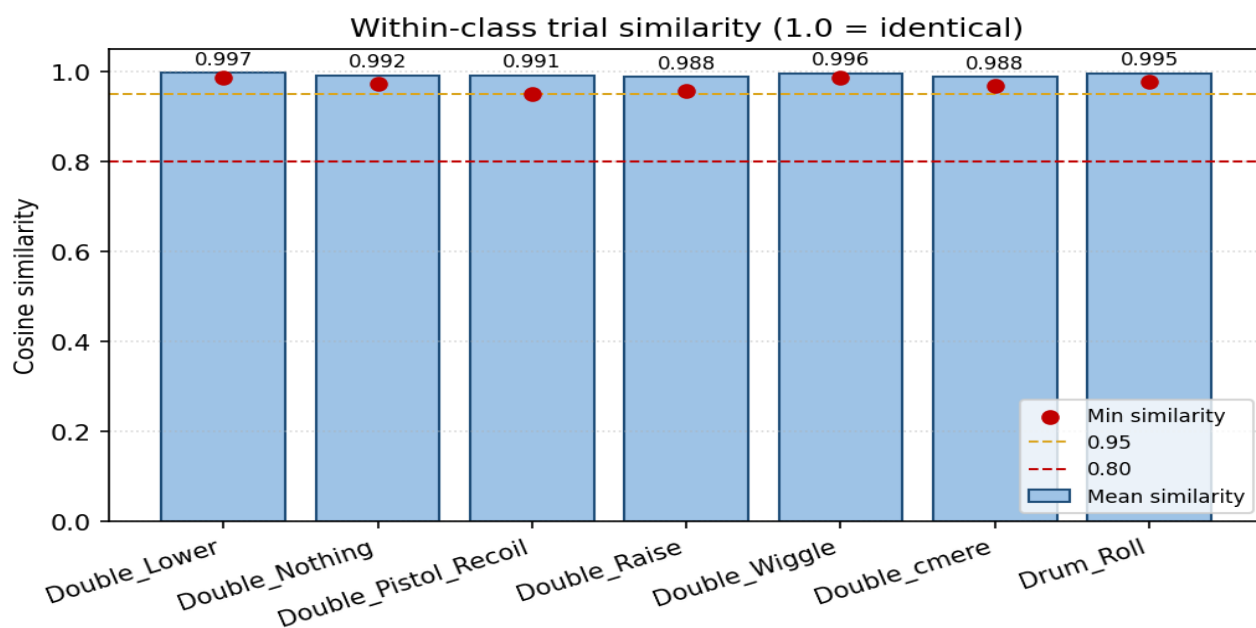


Mean training curves across LOUO folds

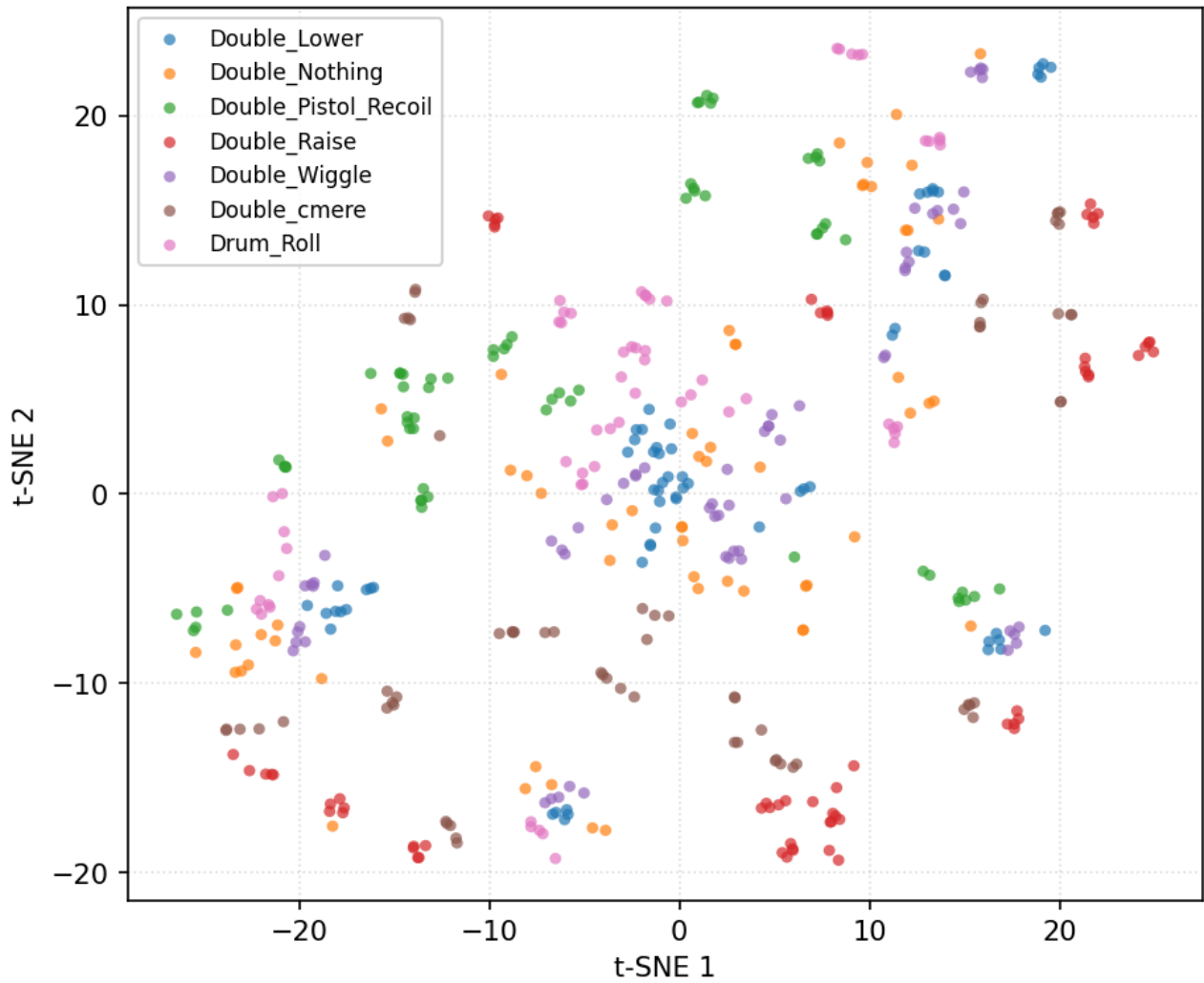


Data diagnostics — pooled across users

Filtered + resampled trials from every LOUO user (before per-trial normalisation).



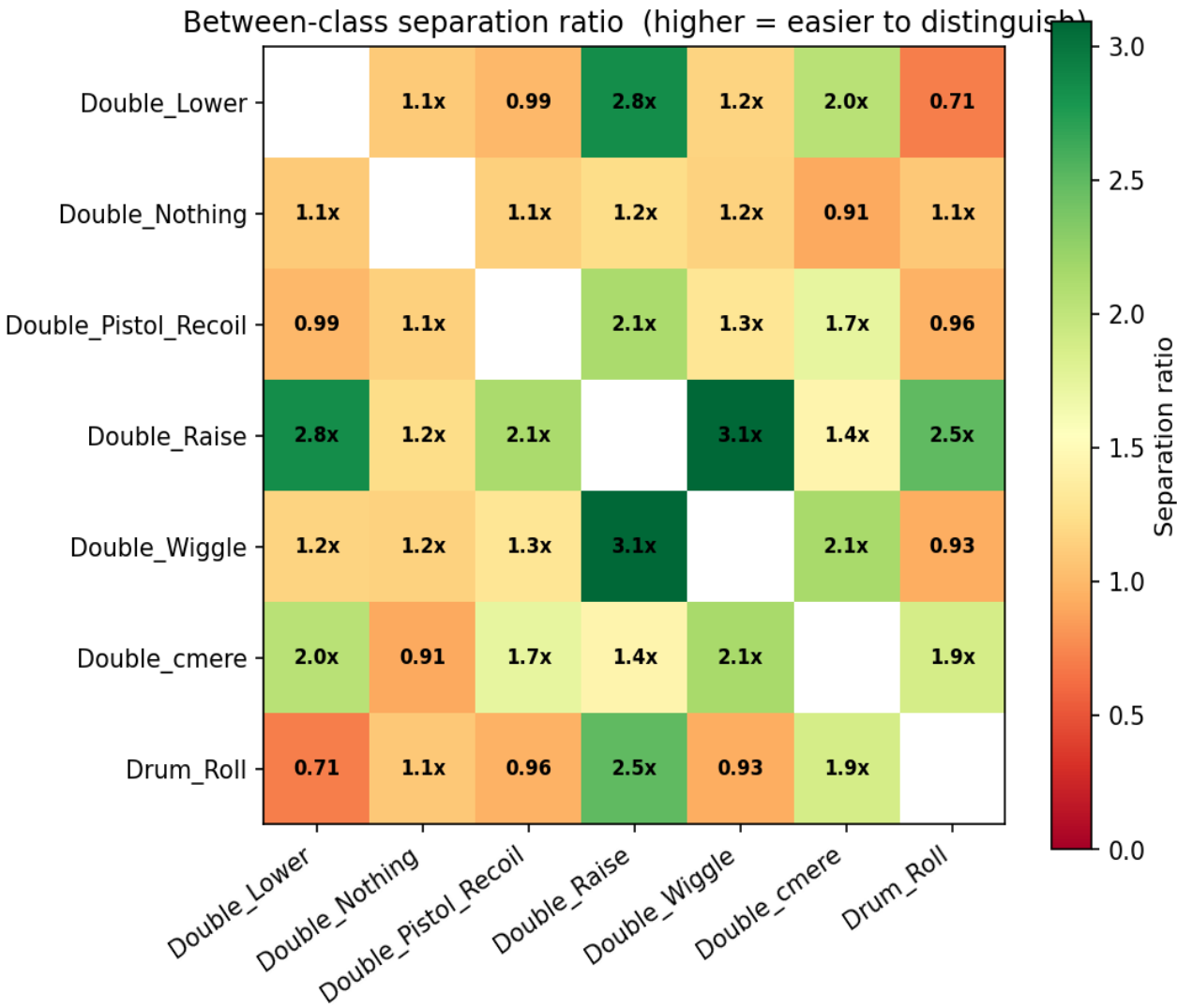
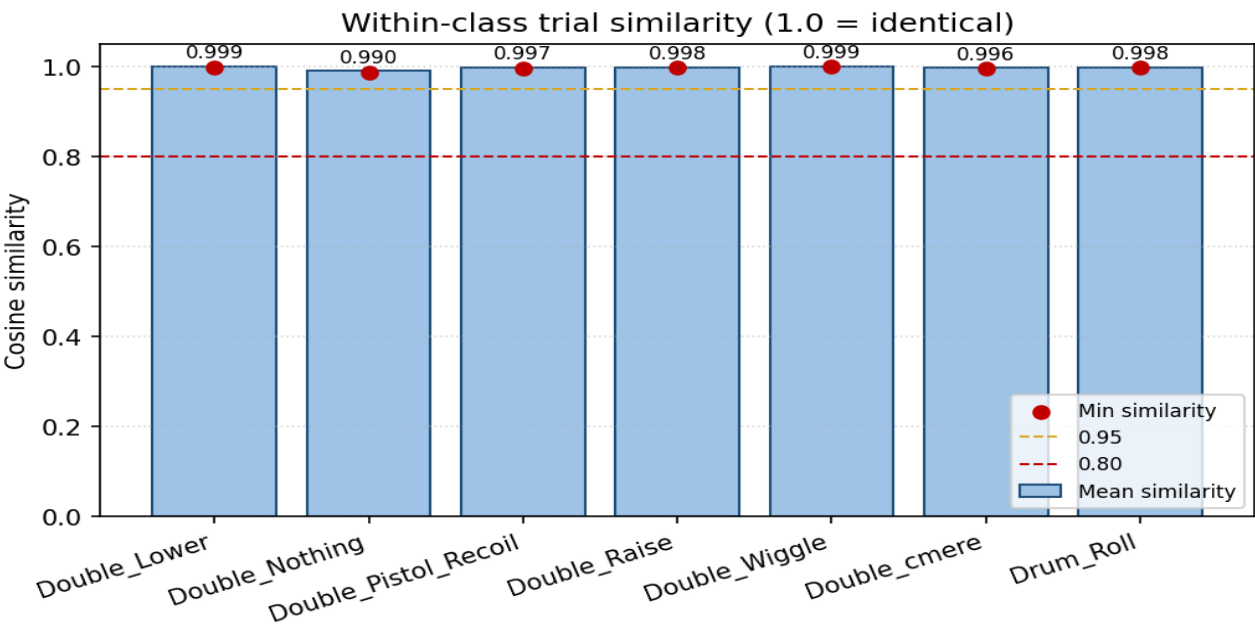
t-SNE feature space projection



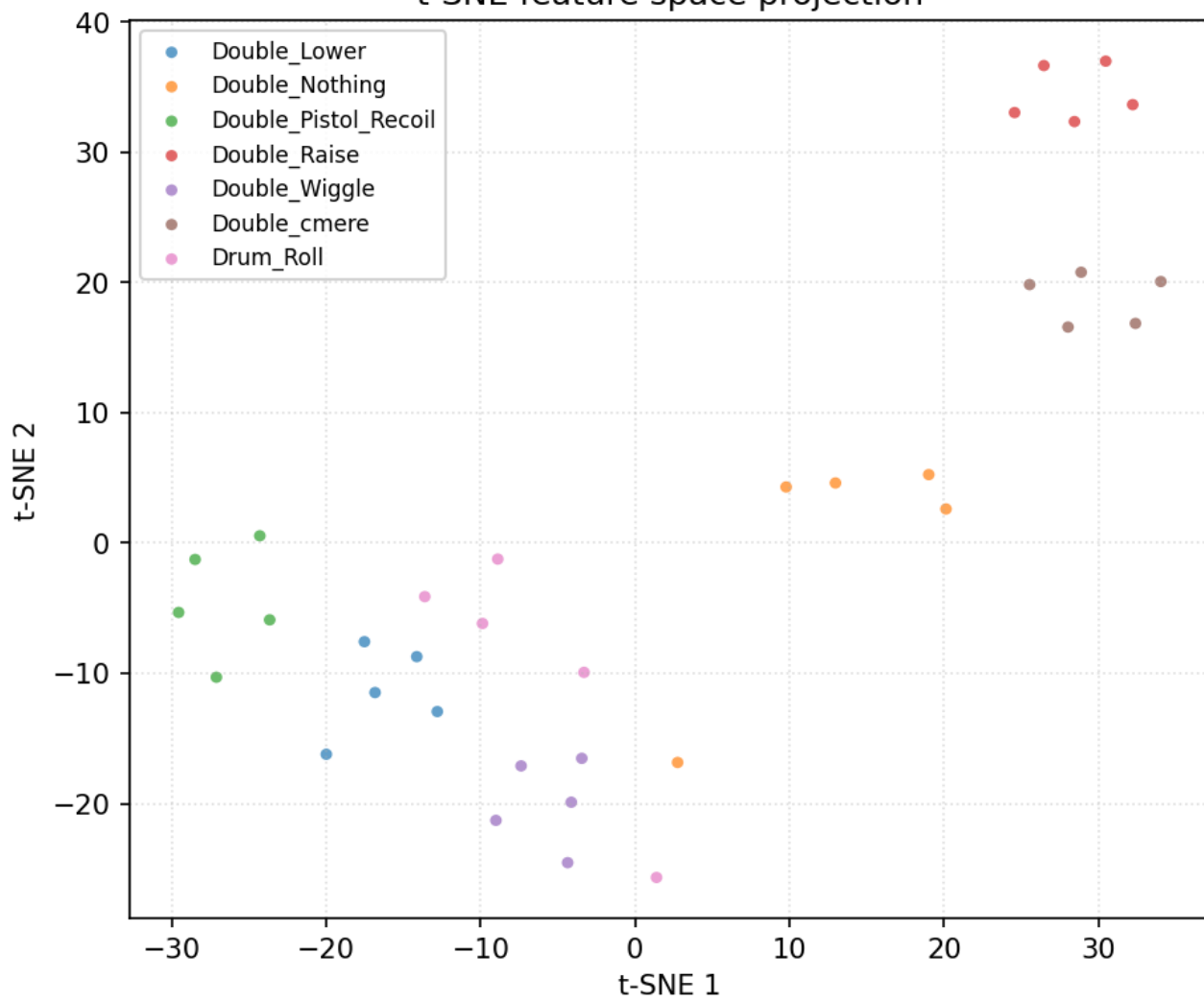
Reading guide: within-class similarity > 0.95 = very consistent; < 0.80 = high variance. Separation ratio > 100x = trivially separable, 20-100x = easy, 5-20x = moderate, < 5x = hard (expect confusion).

Data diagnostics — user 11_Mansh

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

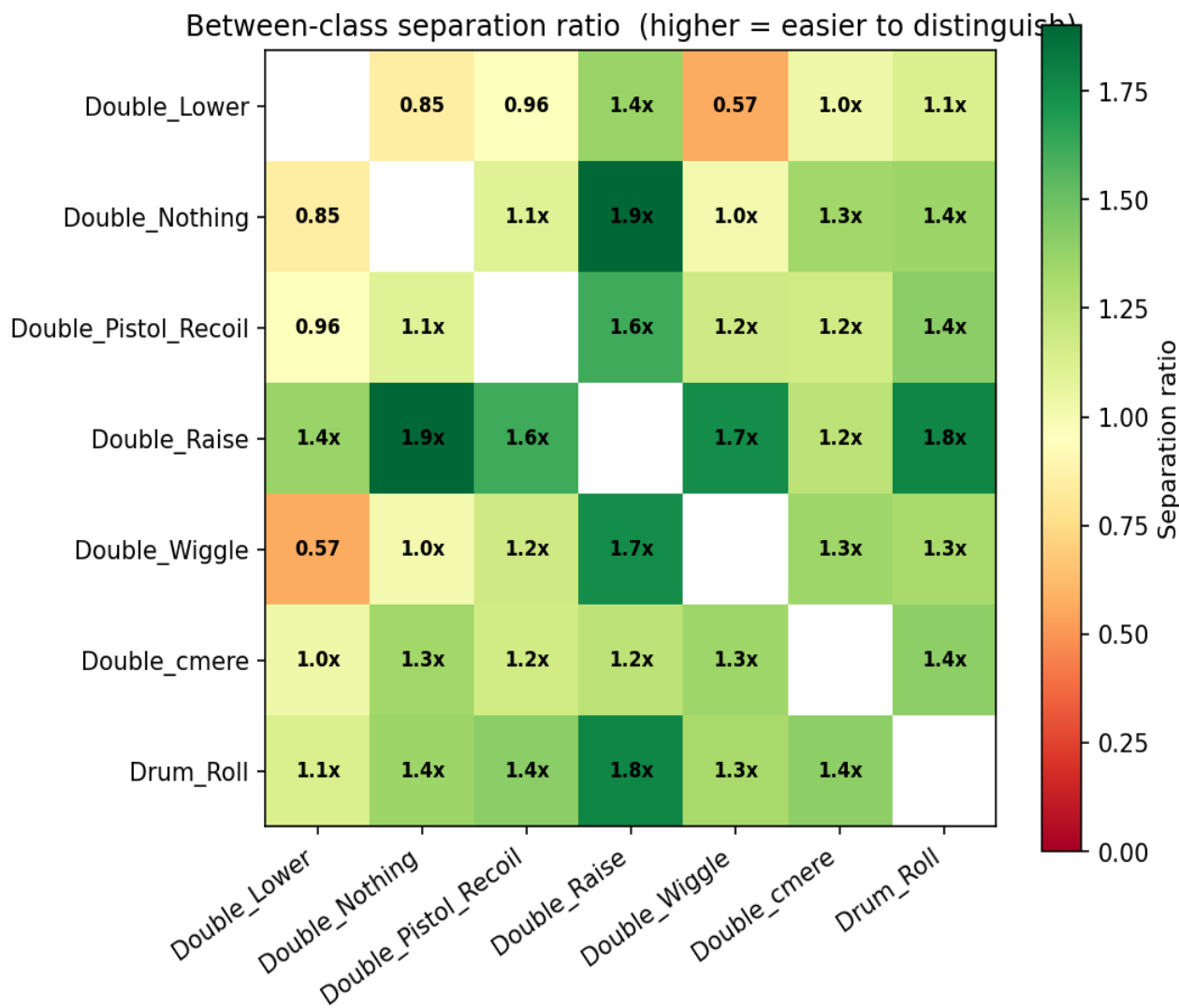
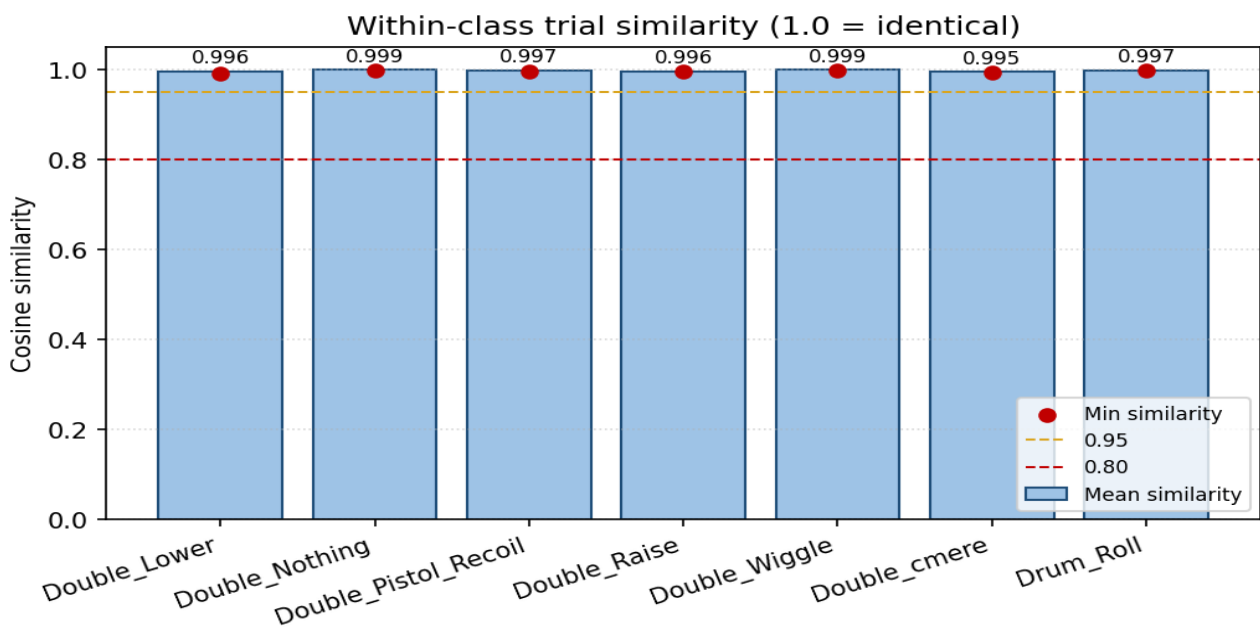


t-SNE feature space projection

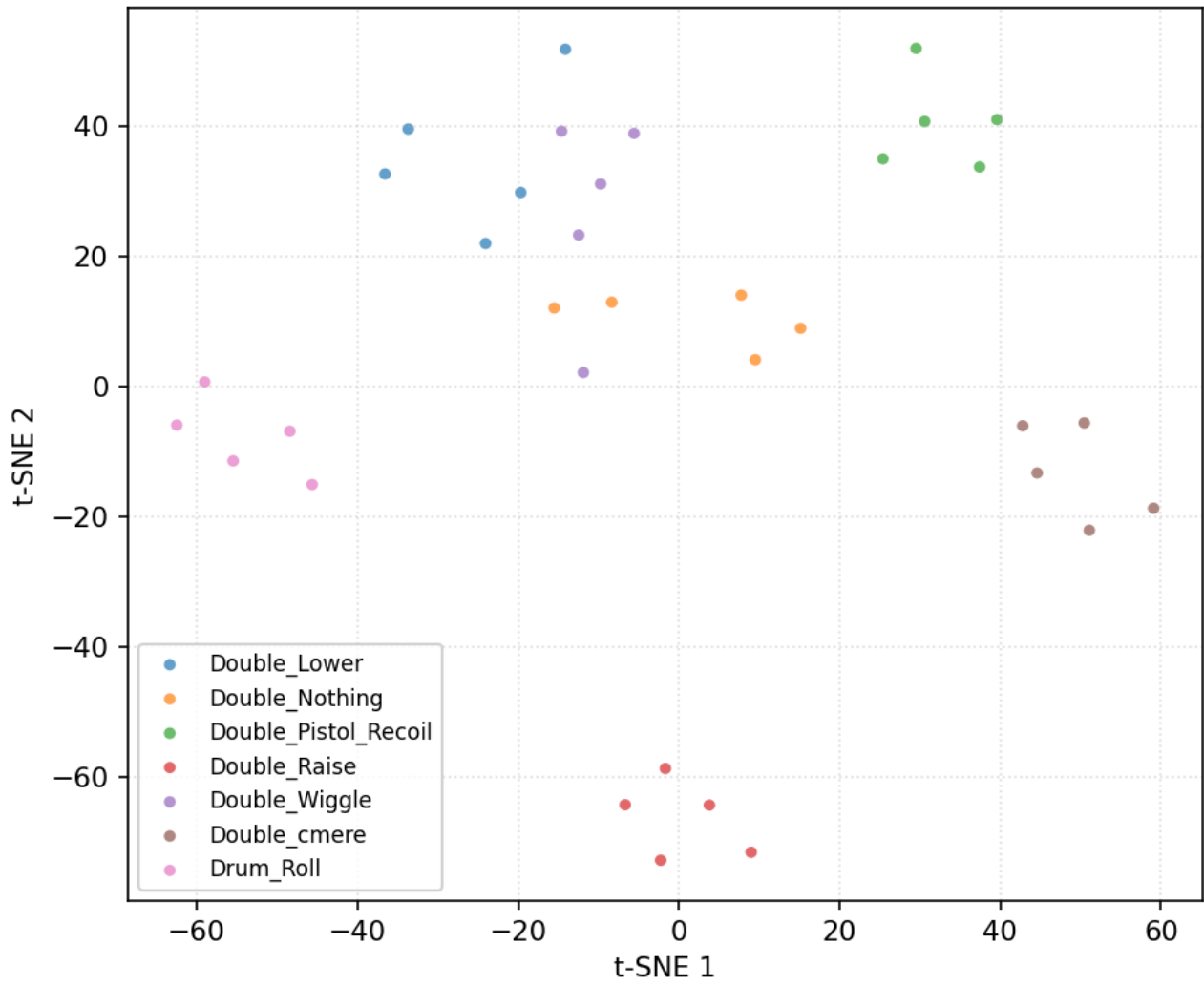


Data diagnostics — user 12_Marcus

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

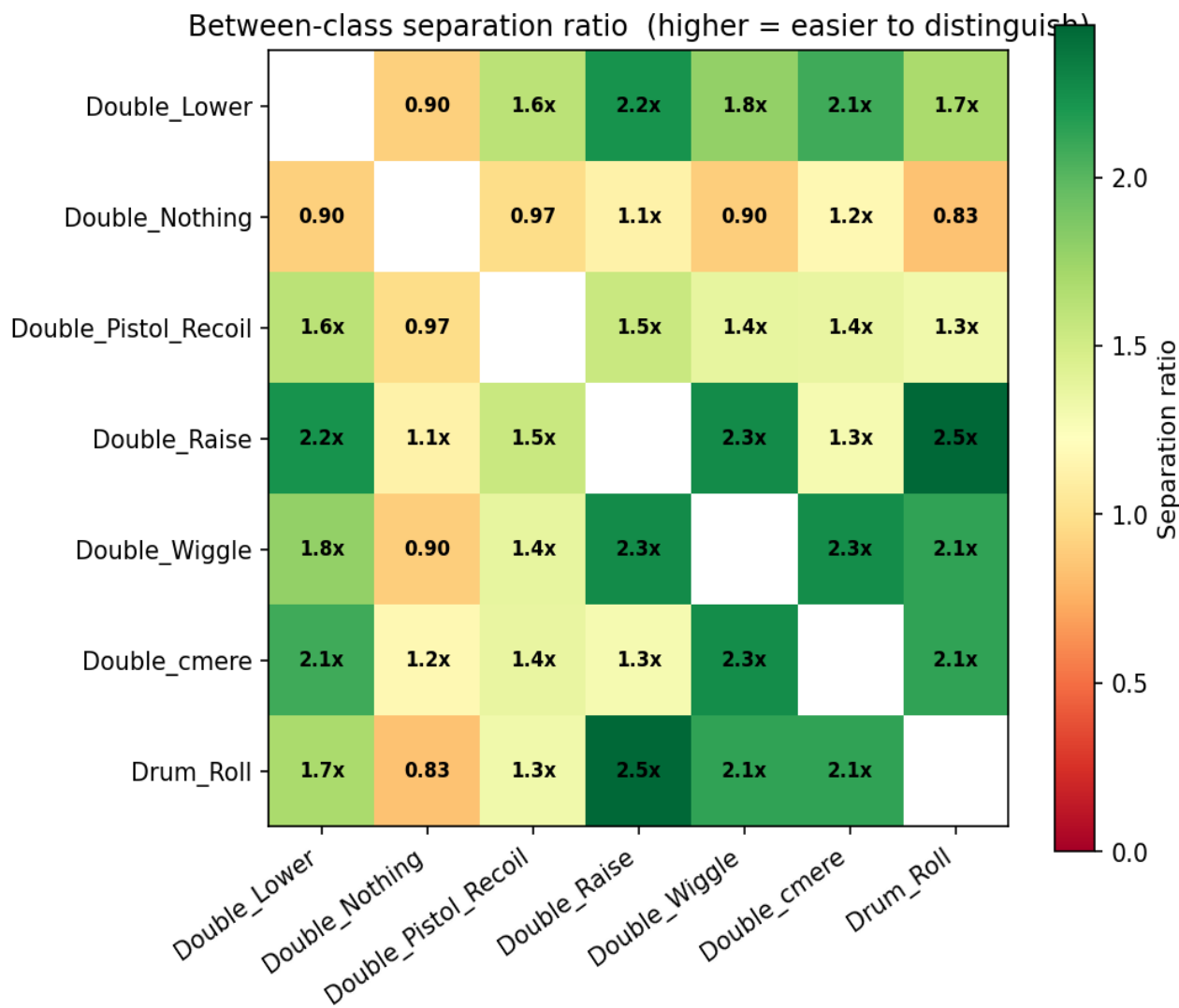
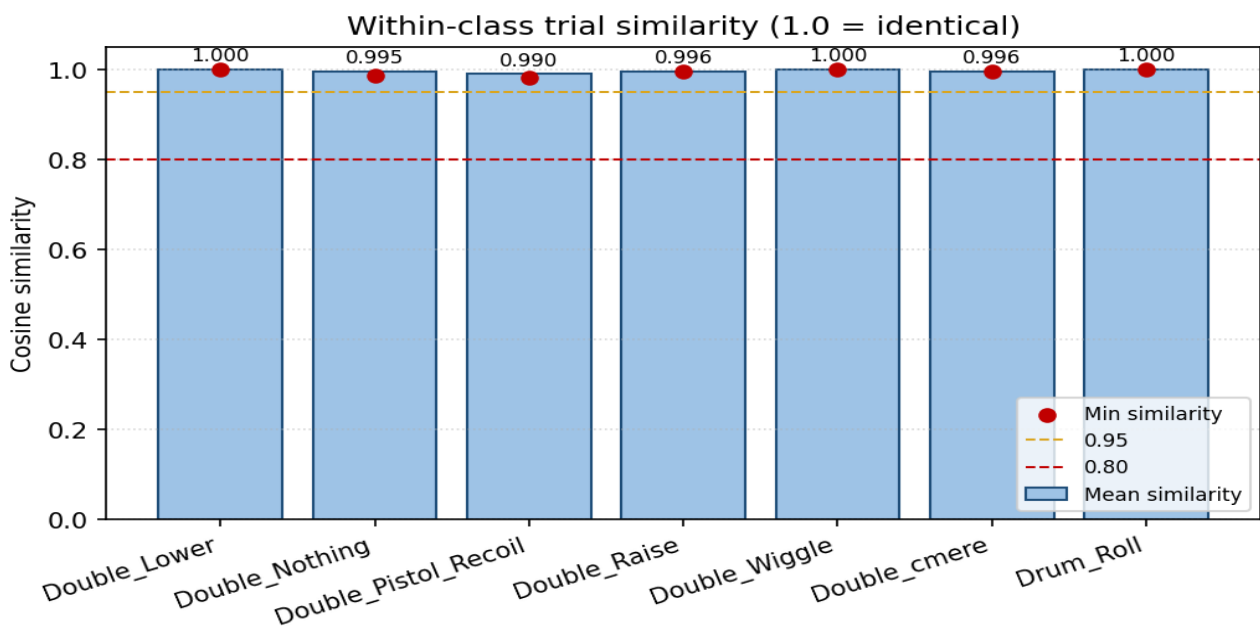


t-SNE feature space projection

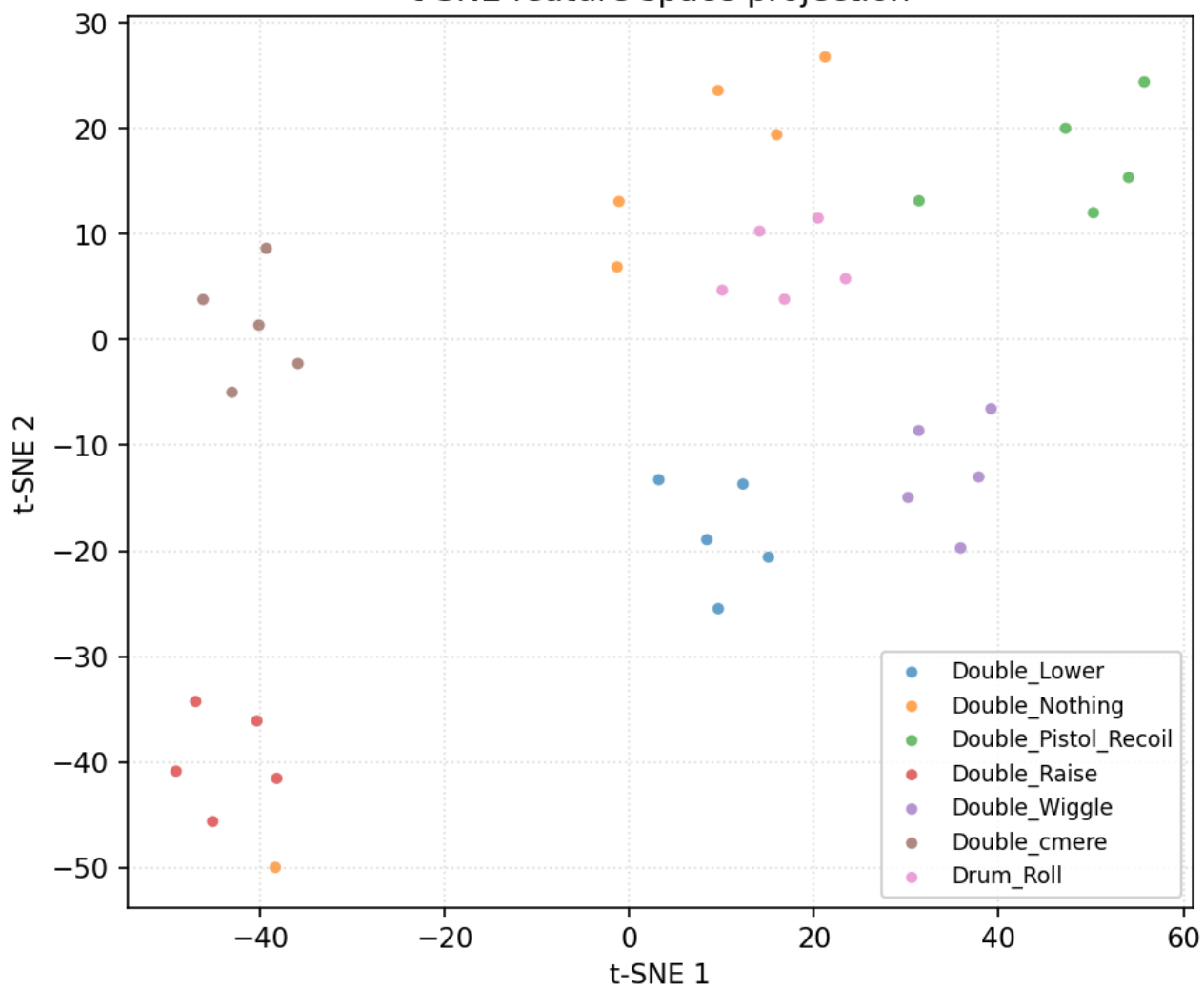


Data diagnostics — user 14_StephenV2

36 trials, 7 classes (filtered + resampled, before per-trial normalisation).

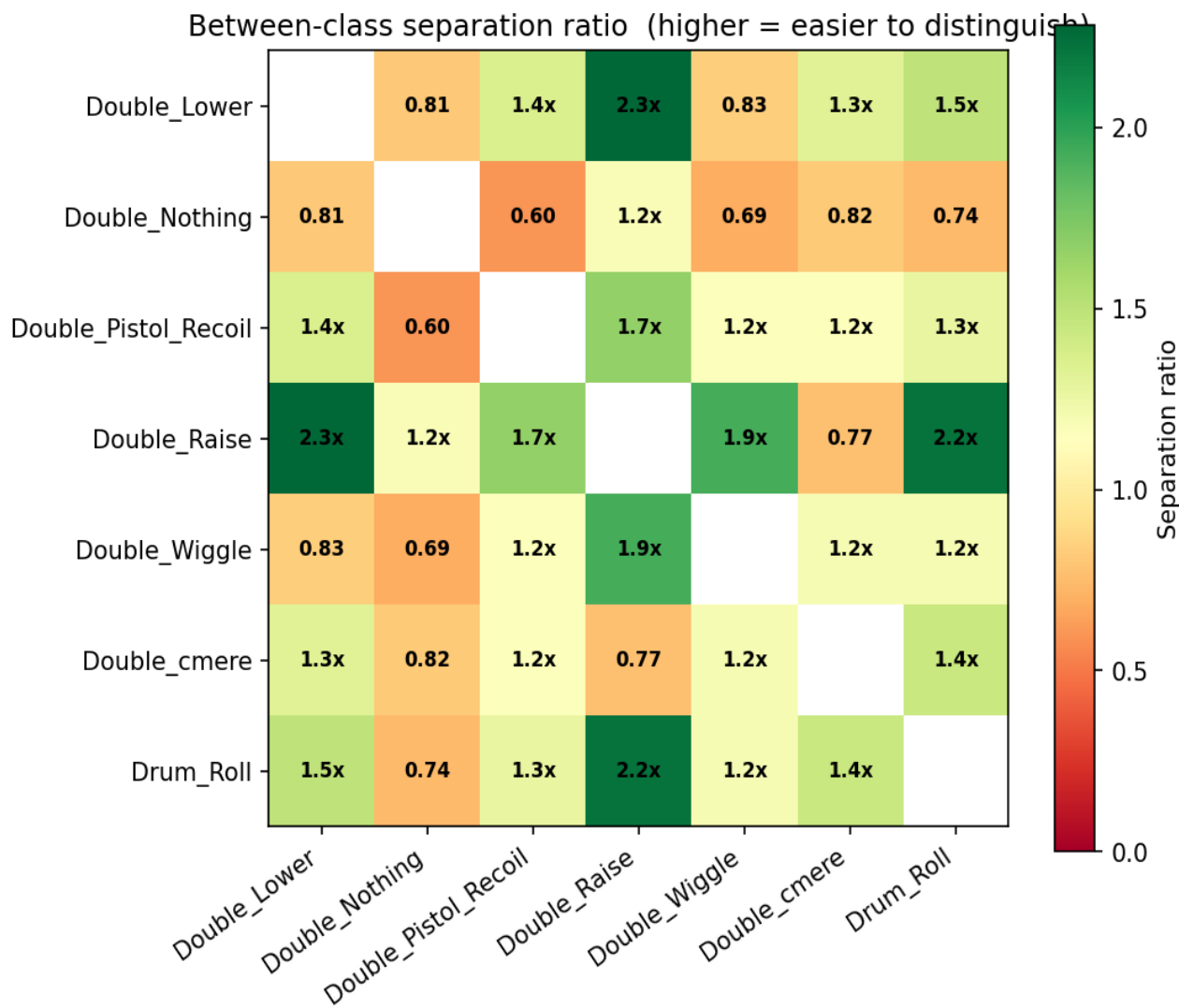
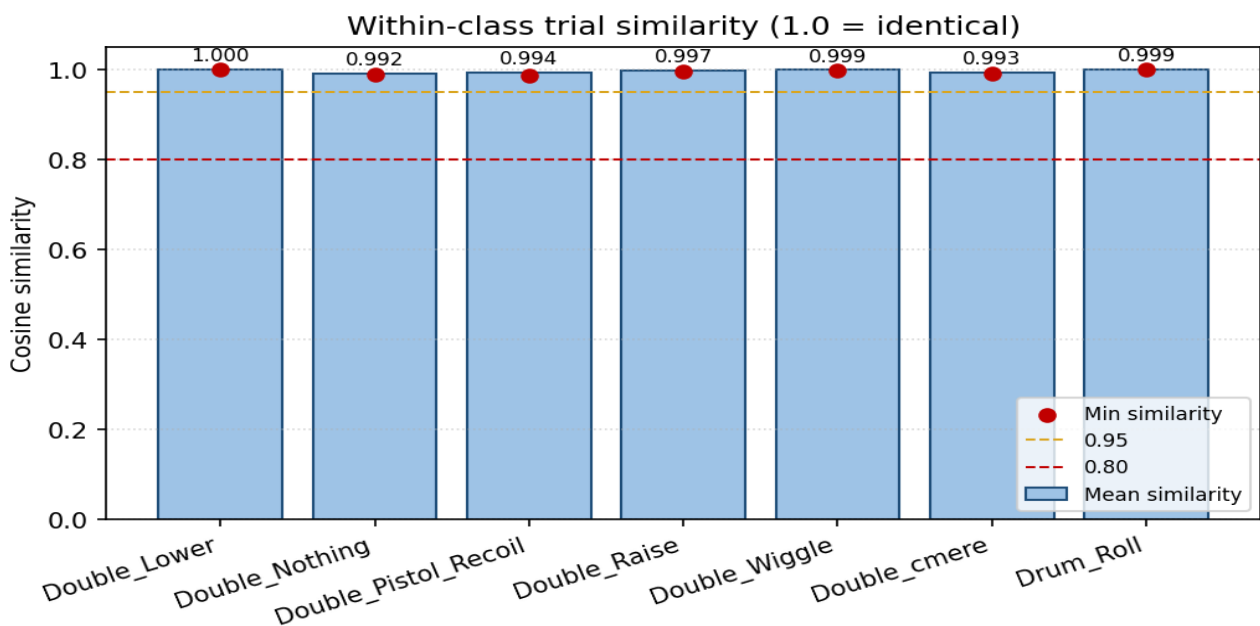


t-SNE feature space projection

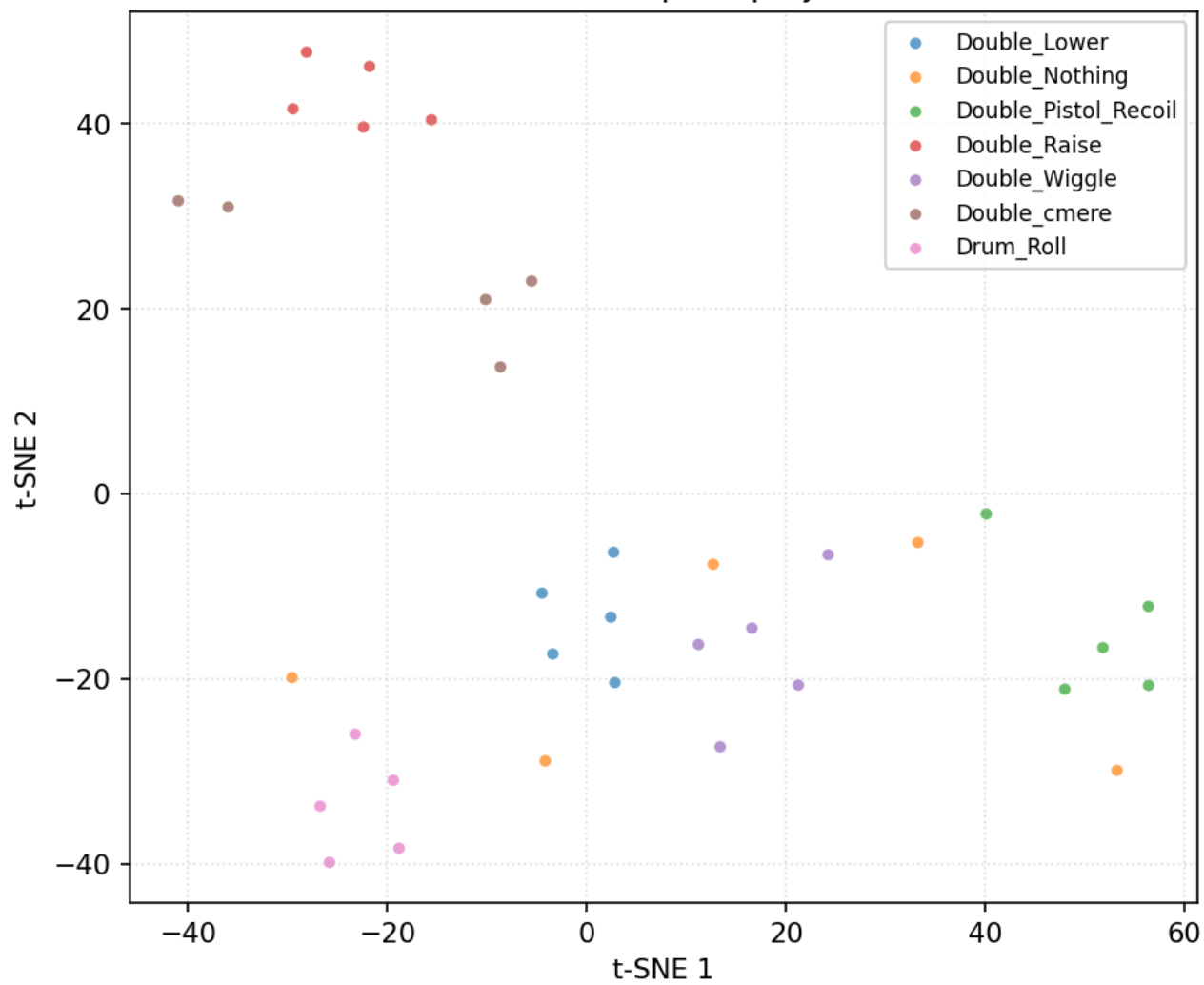


Data diagnostics — user 15_Tash

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

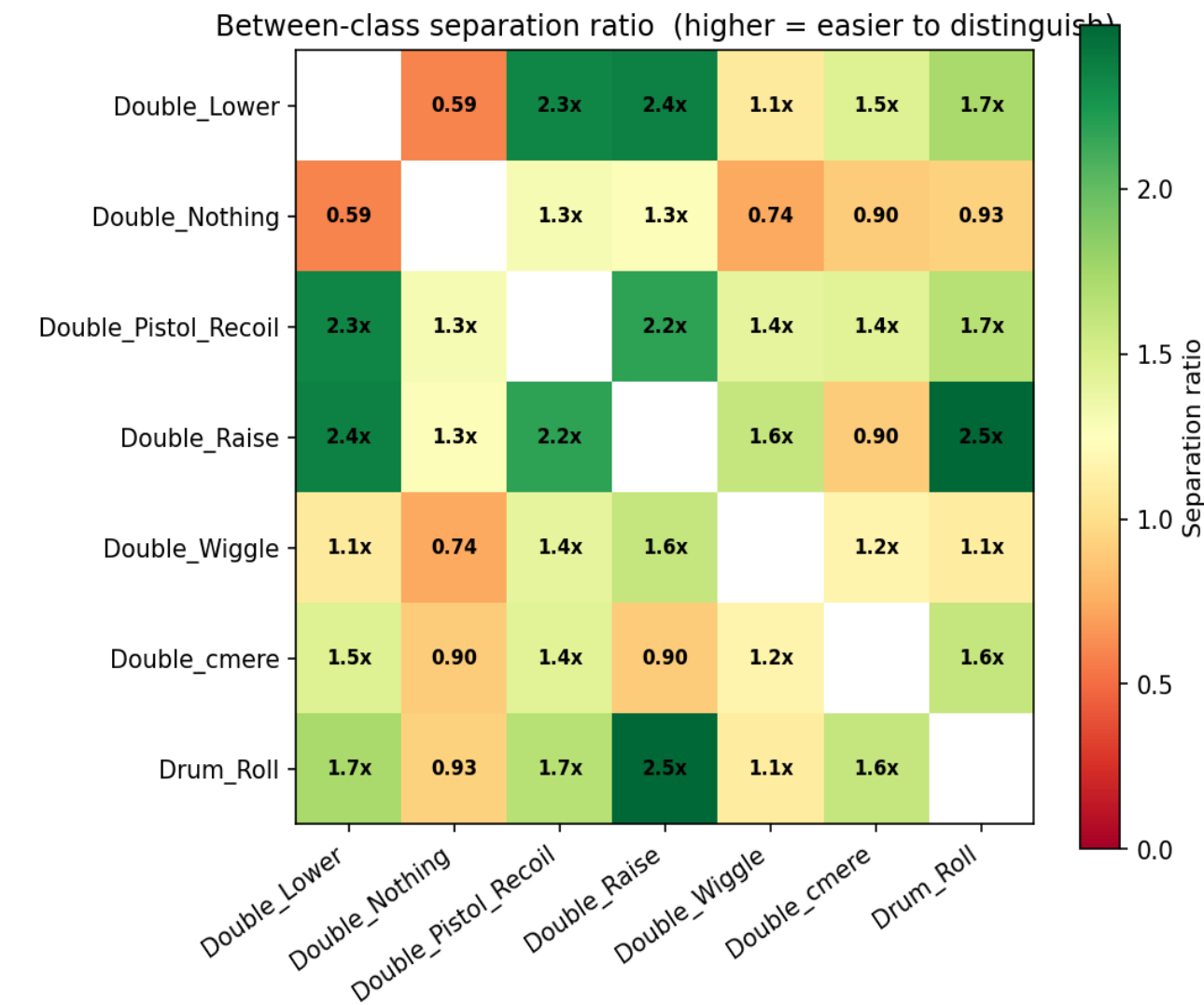
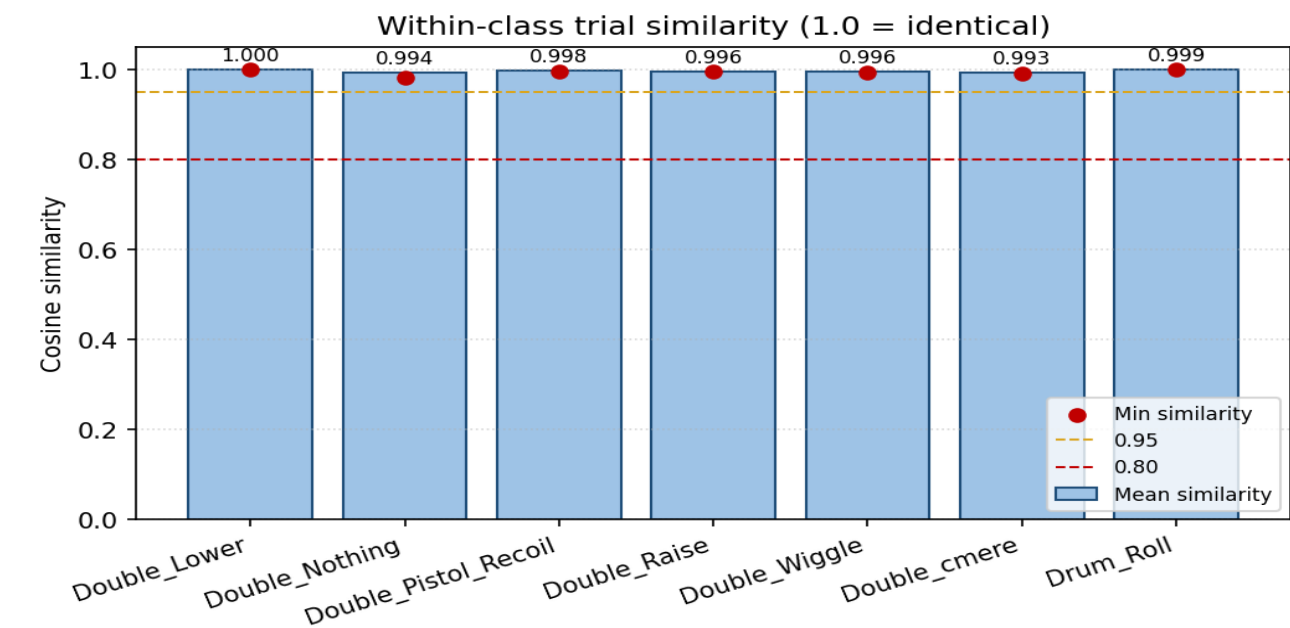


t-SNE feature space projection

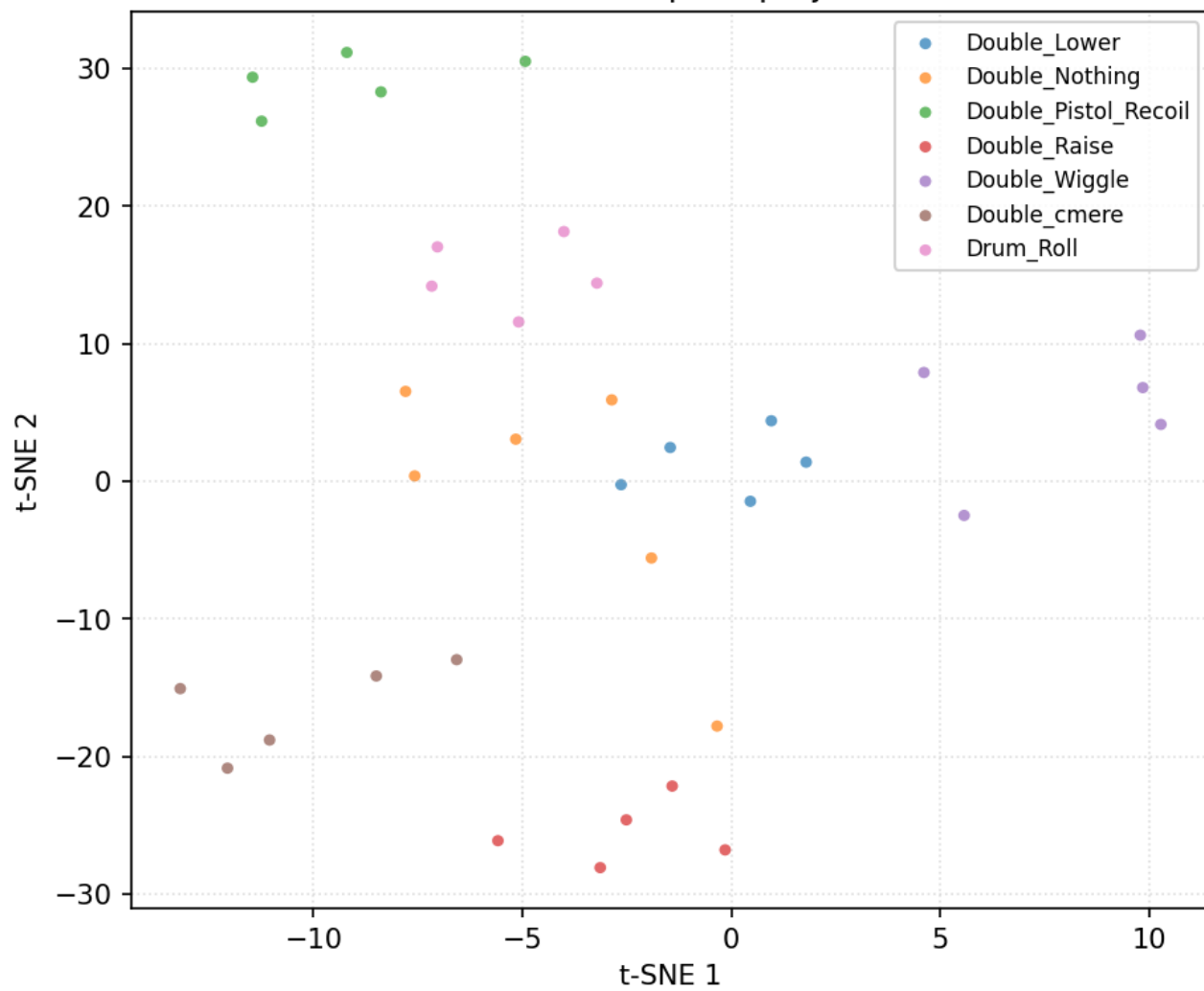


Data diagnostics — user 16_Bella

36 trials, 7 classes (filtered + resampled, before per-trial normalisation).

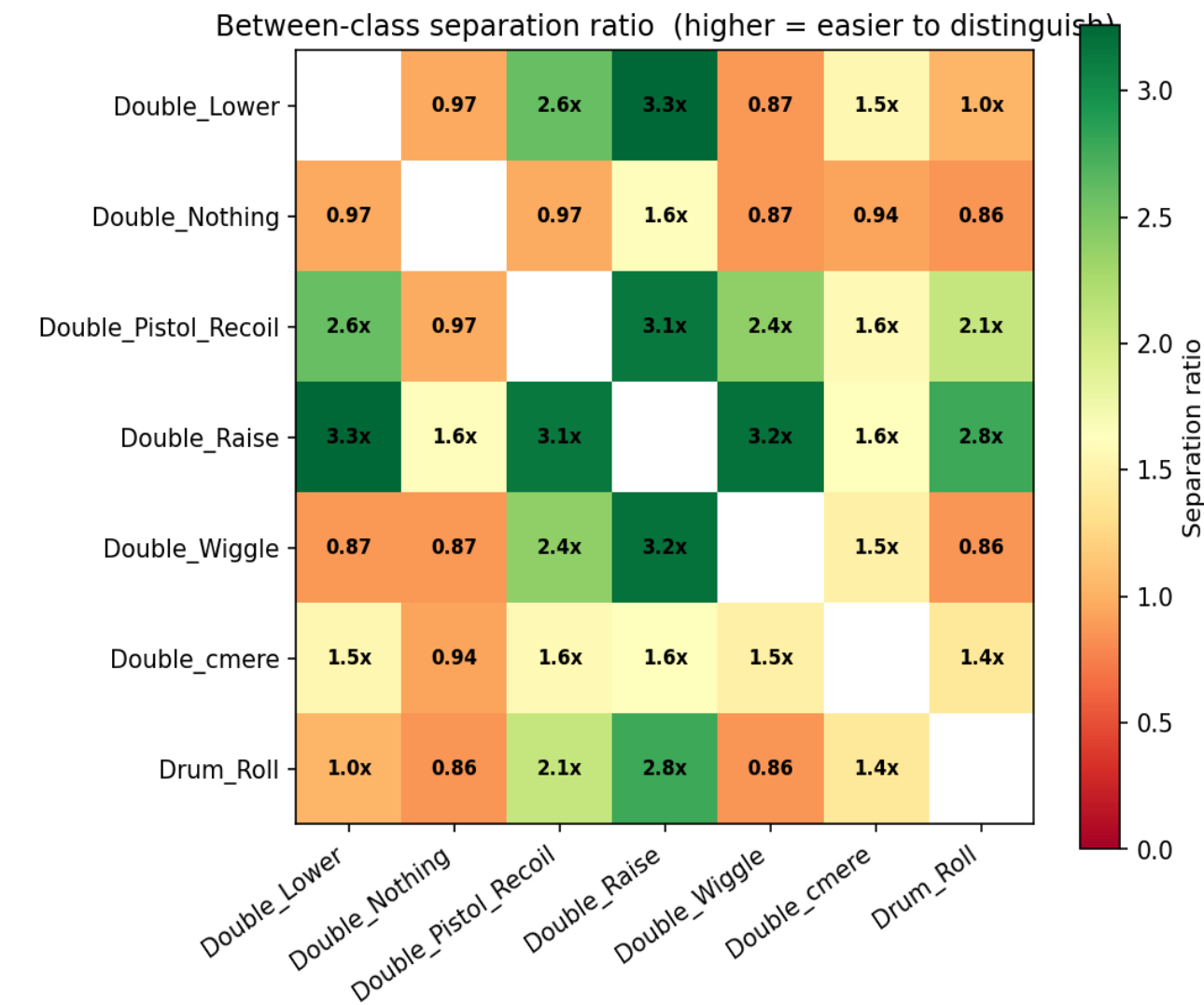
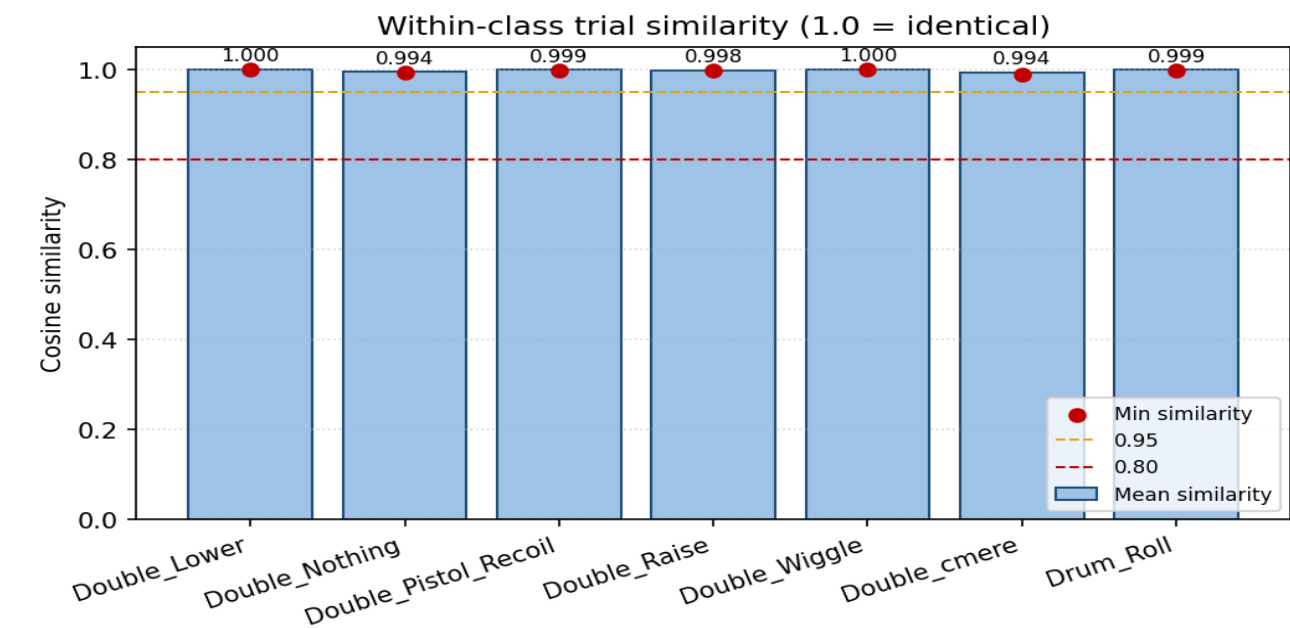


t-SNE feature space projection

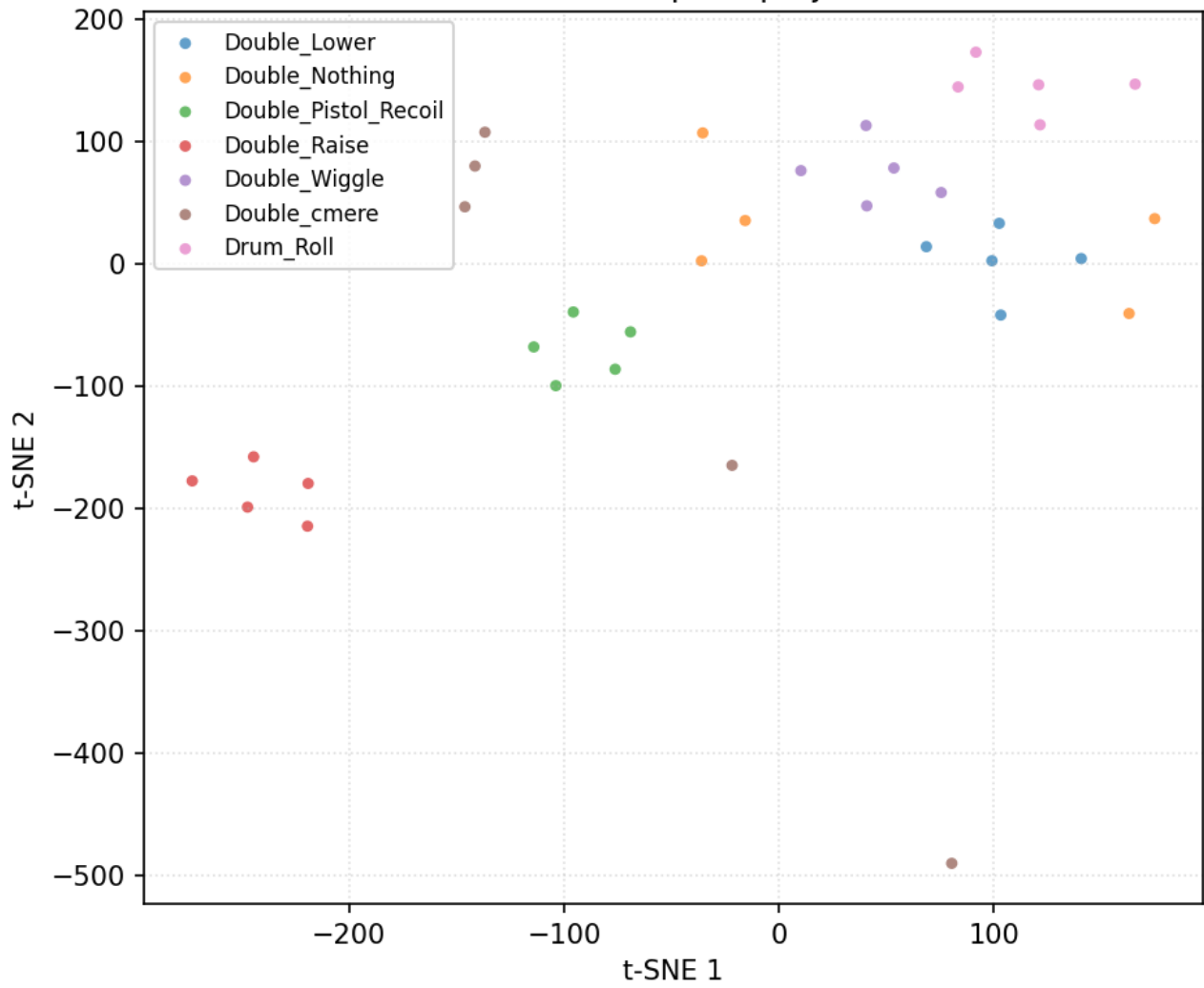


Data diagnostics — user 17_Raquel

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

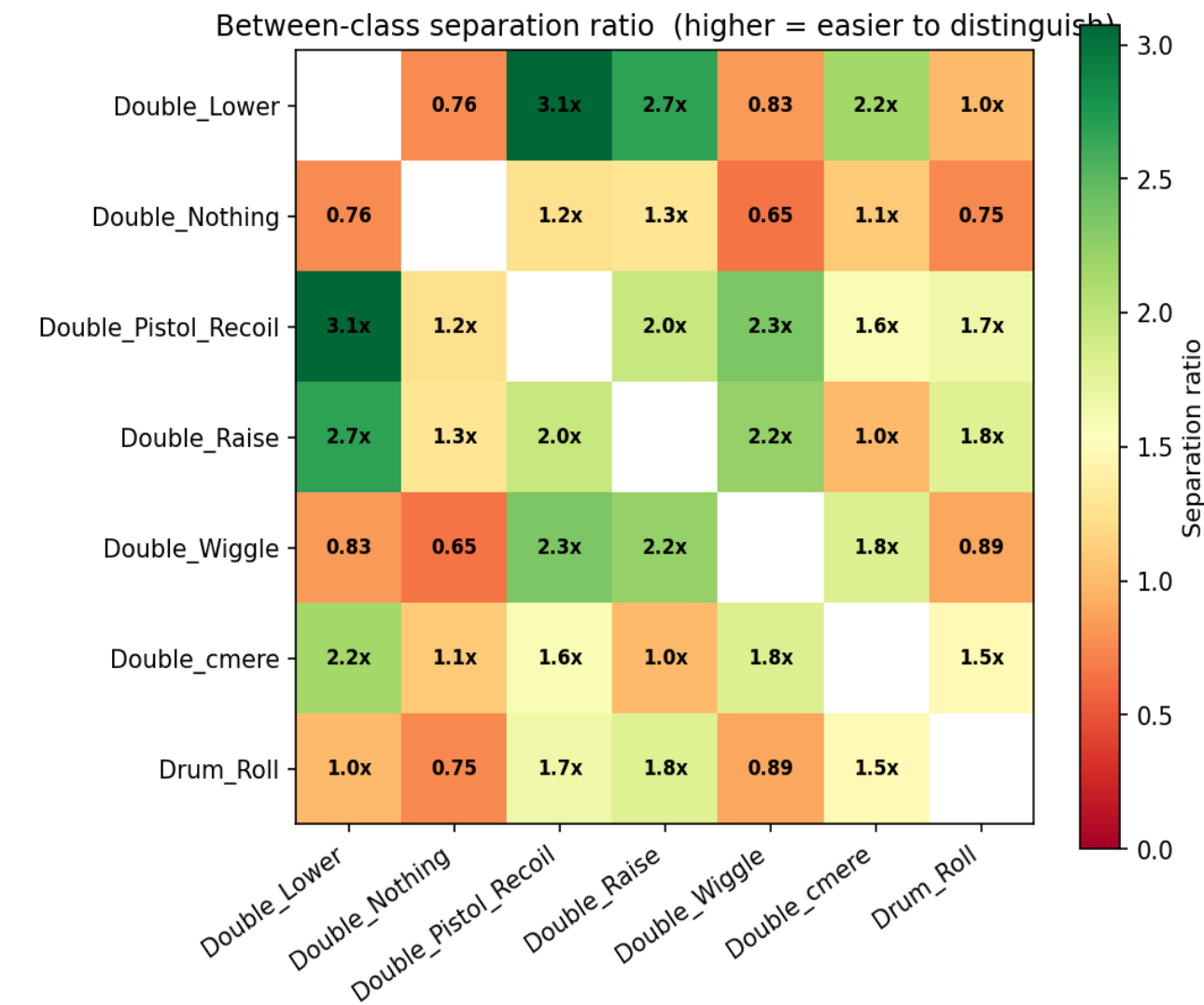
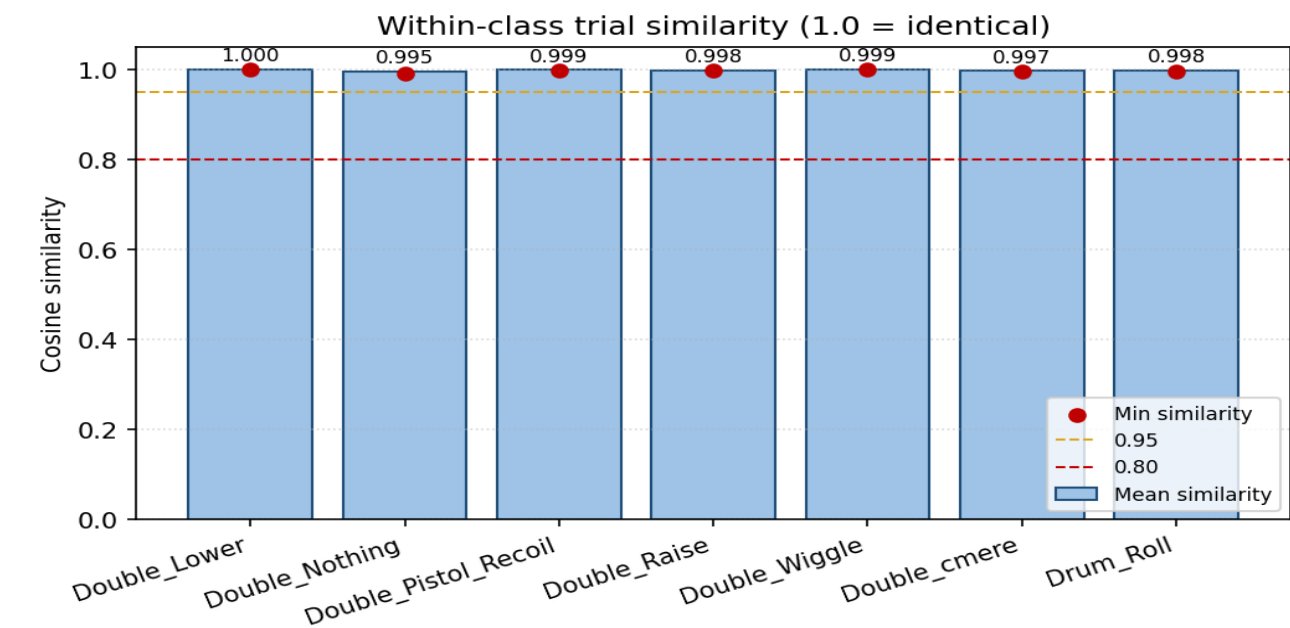


t-SNE feature space projection

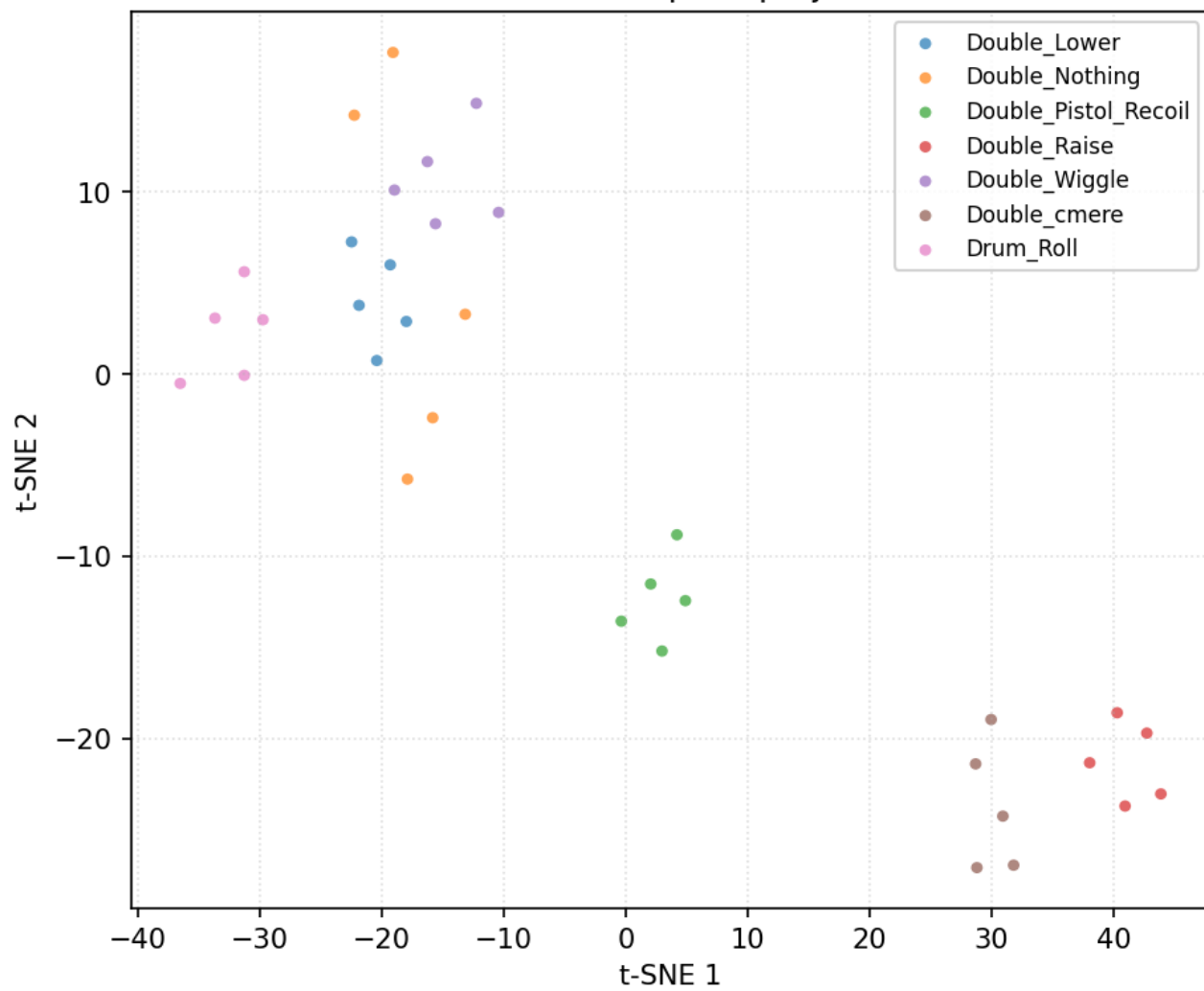


Data diagnostics — user 18_Bridgette

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

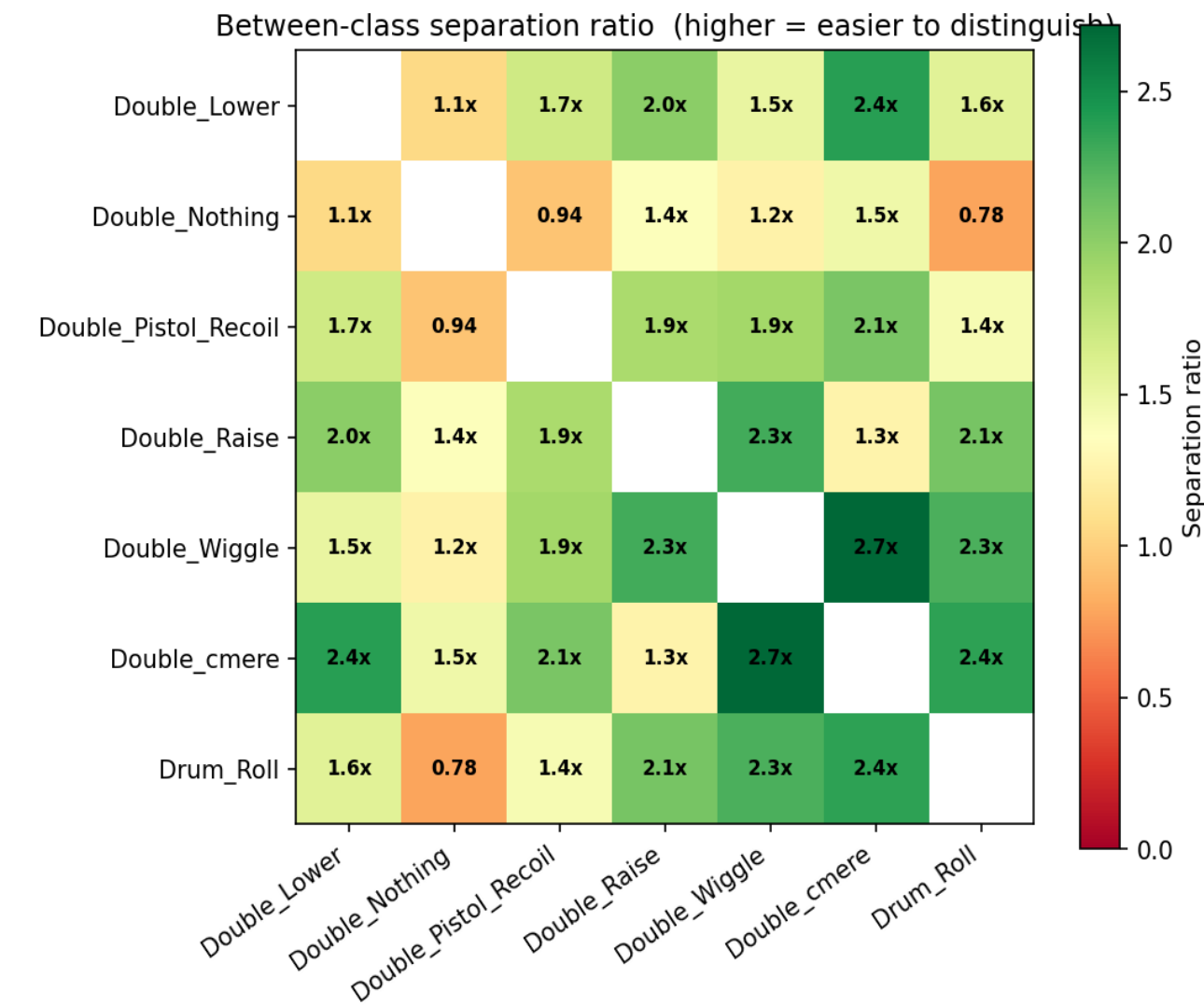
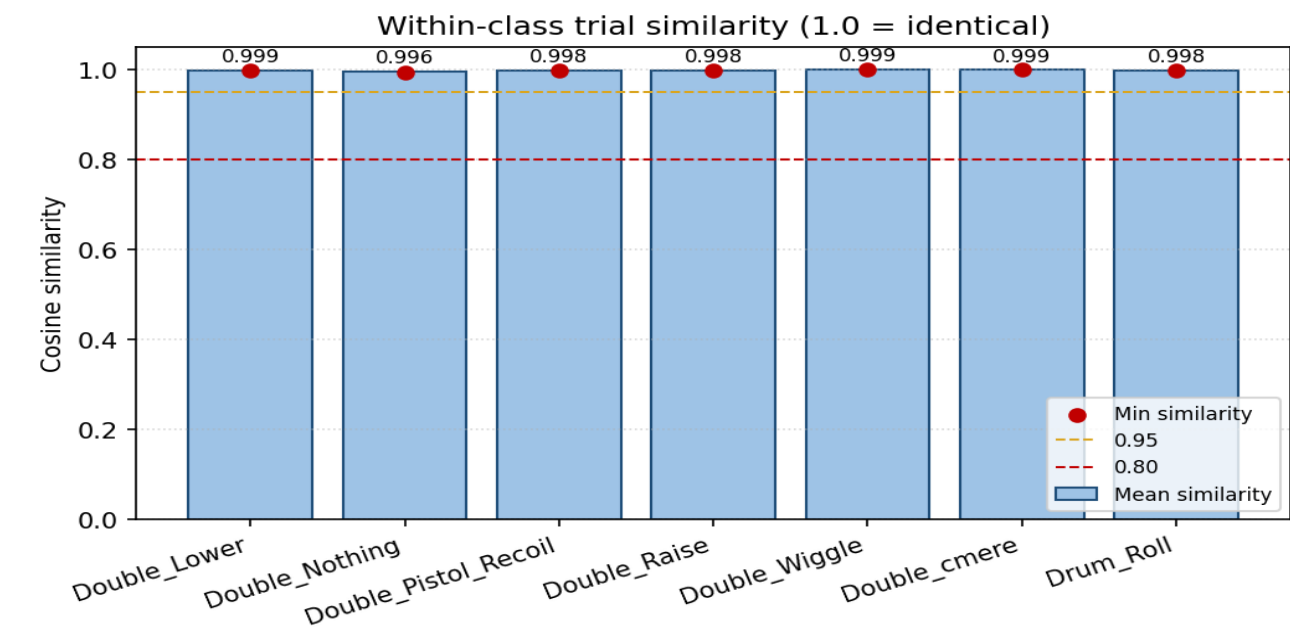


t-SNE feature space projection

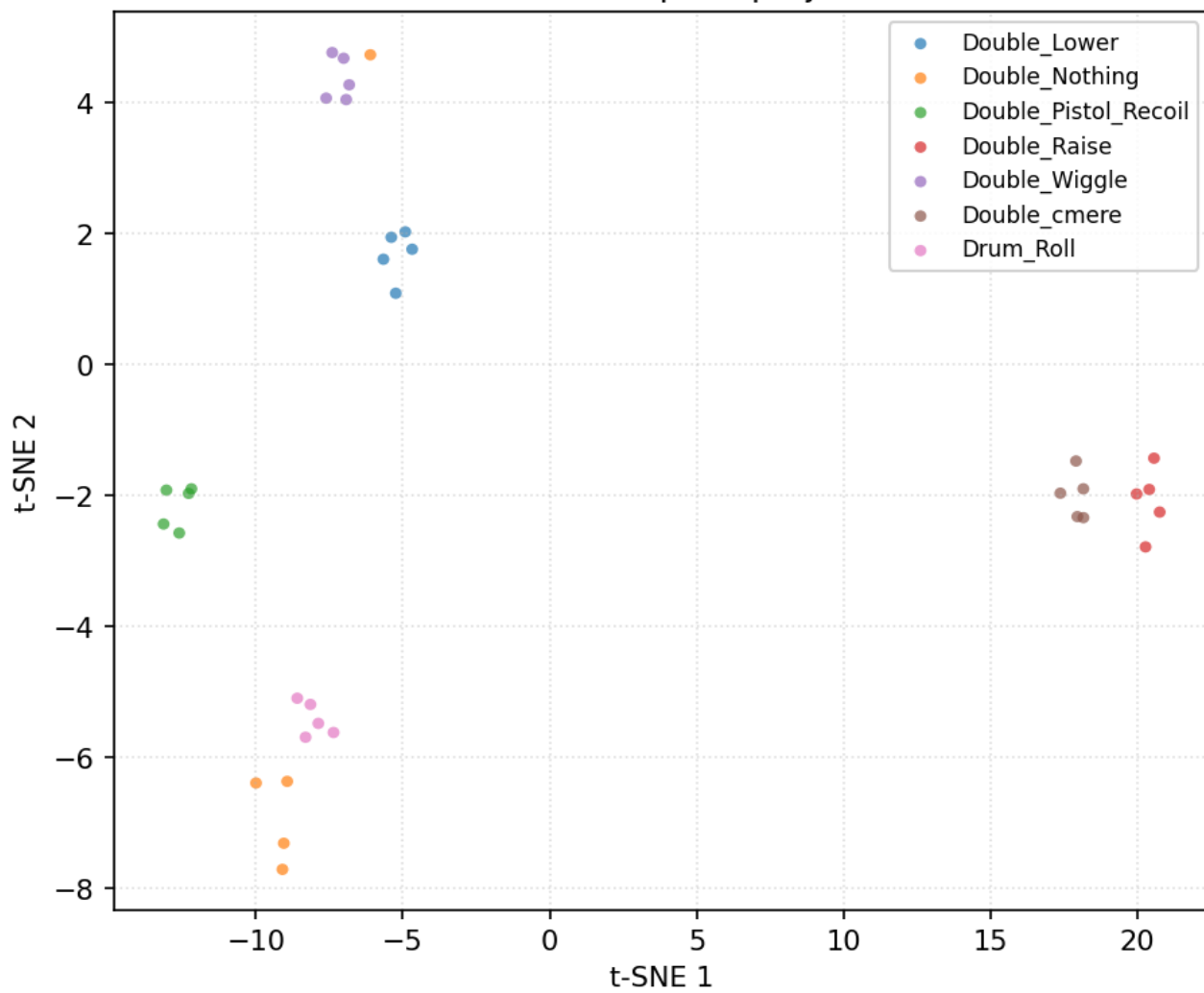


Data diagnostics — user 1_Alán

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

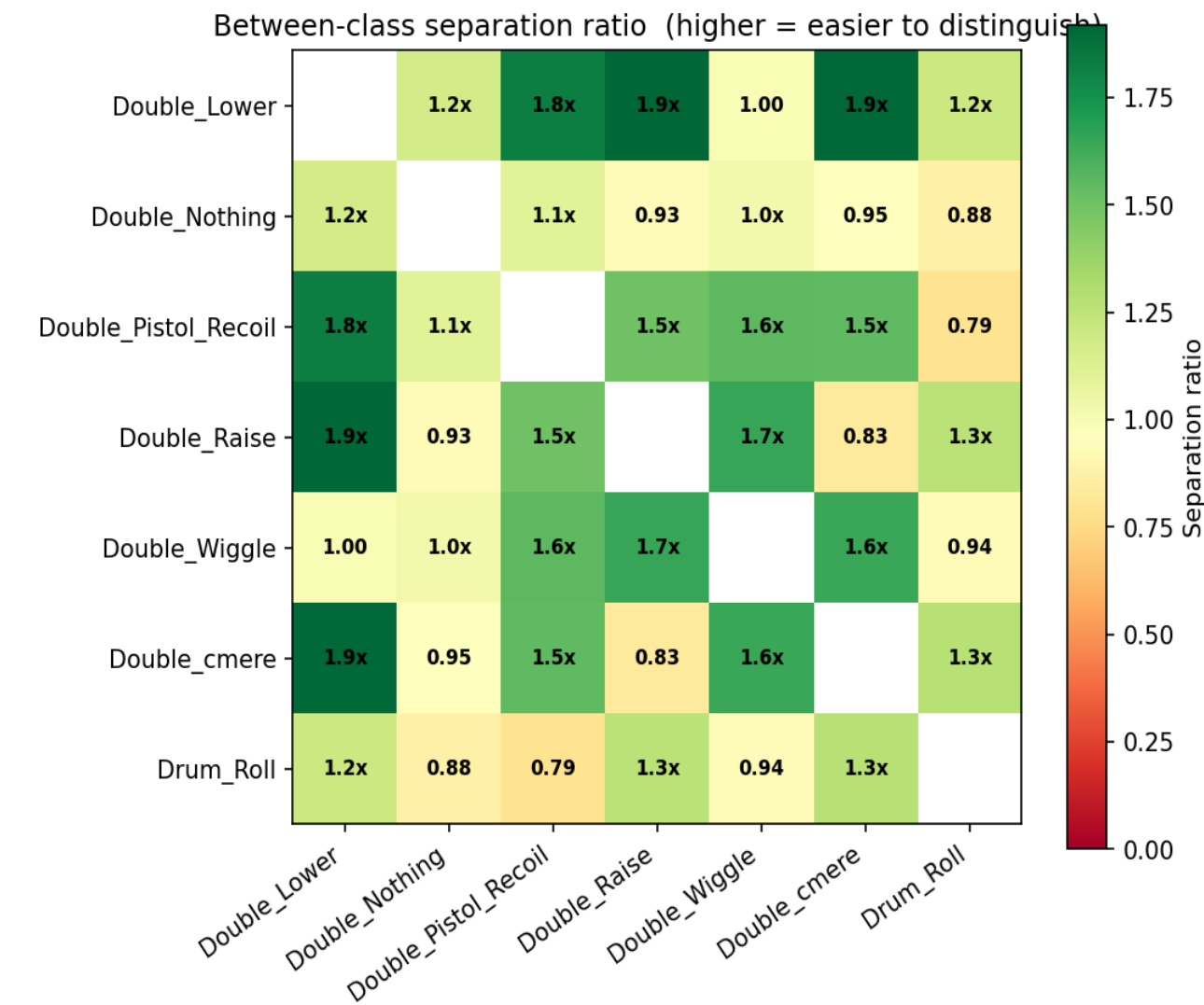
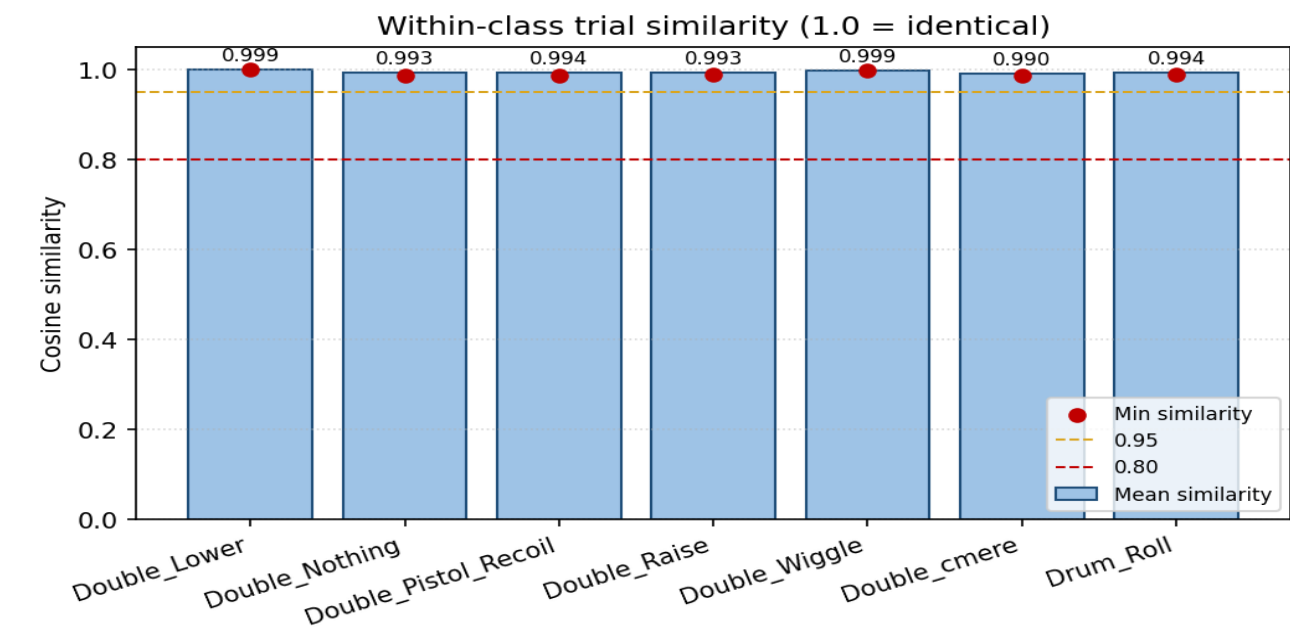


t-SNE feature space projection

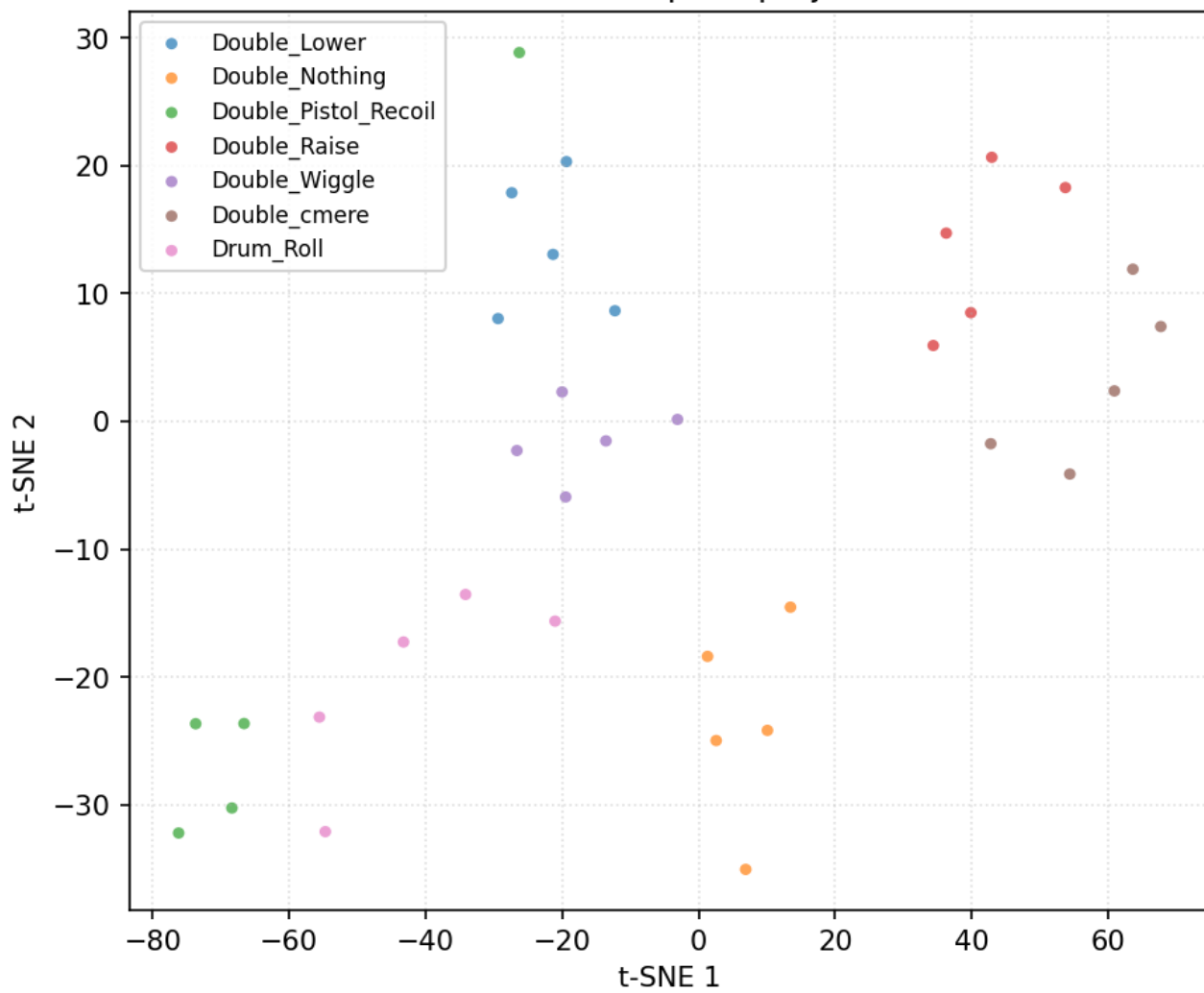


Data diagnostics — user 2_Alex

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

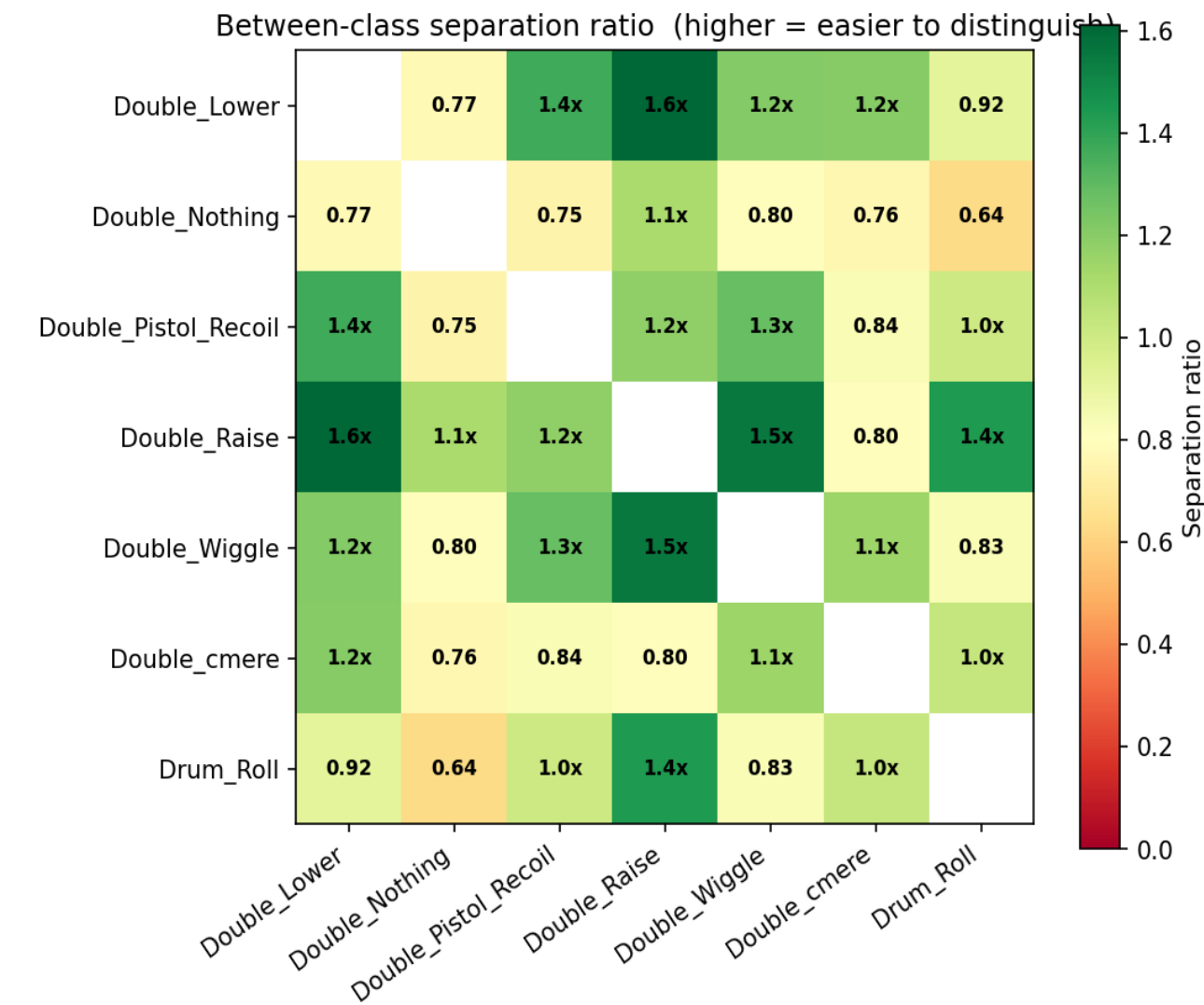
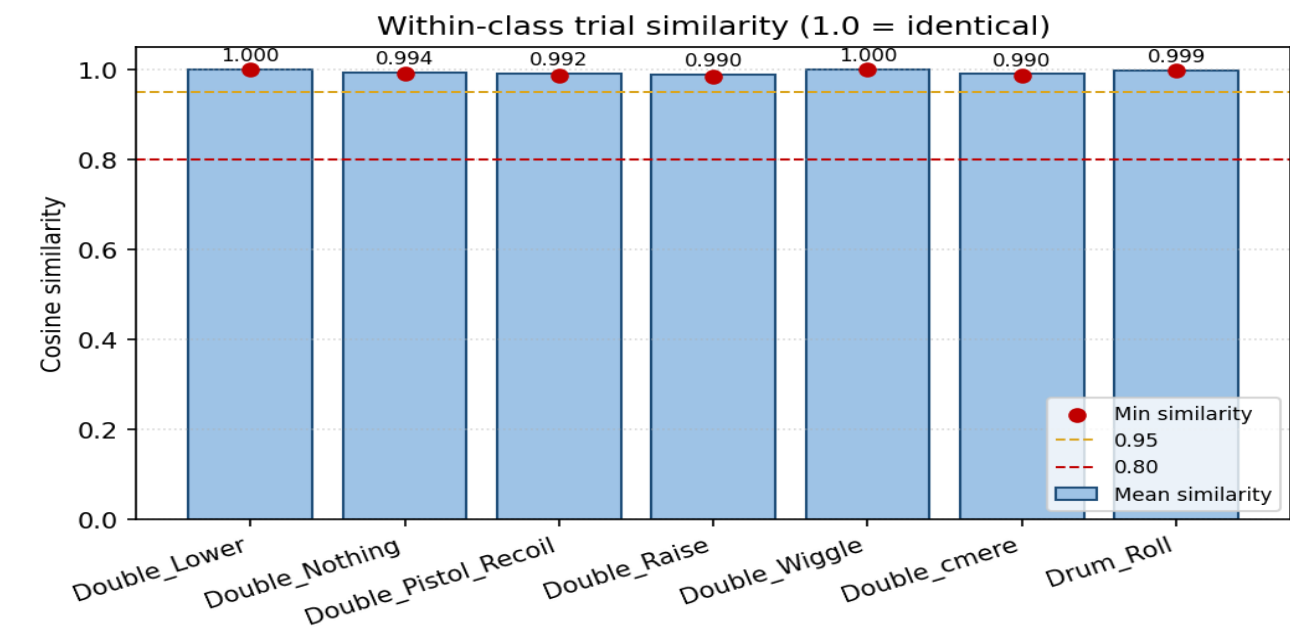


t-SNE feature space projection

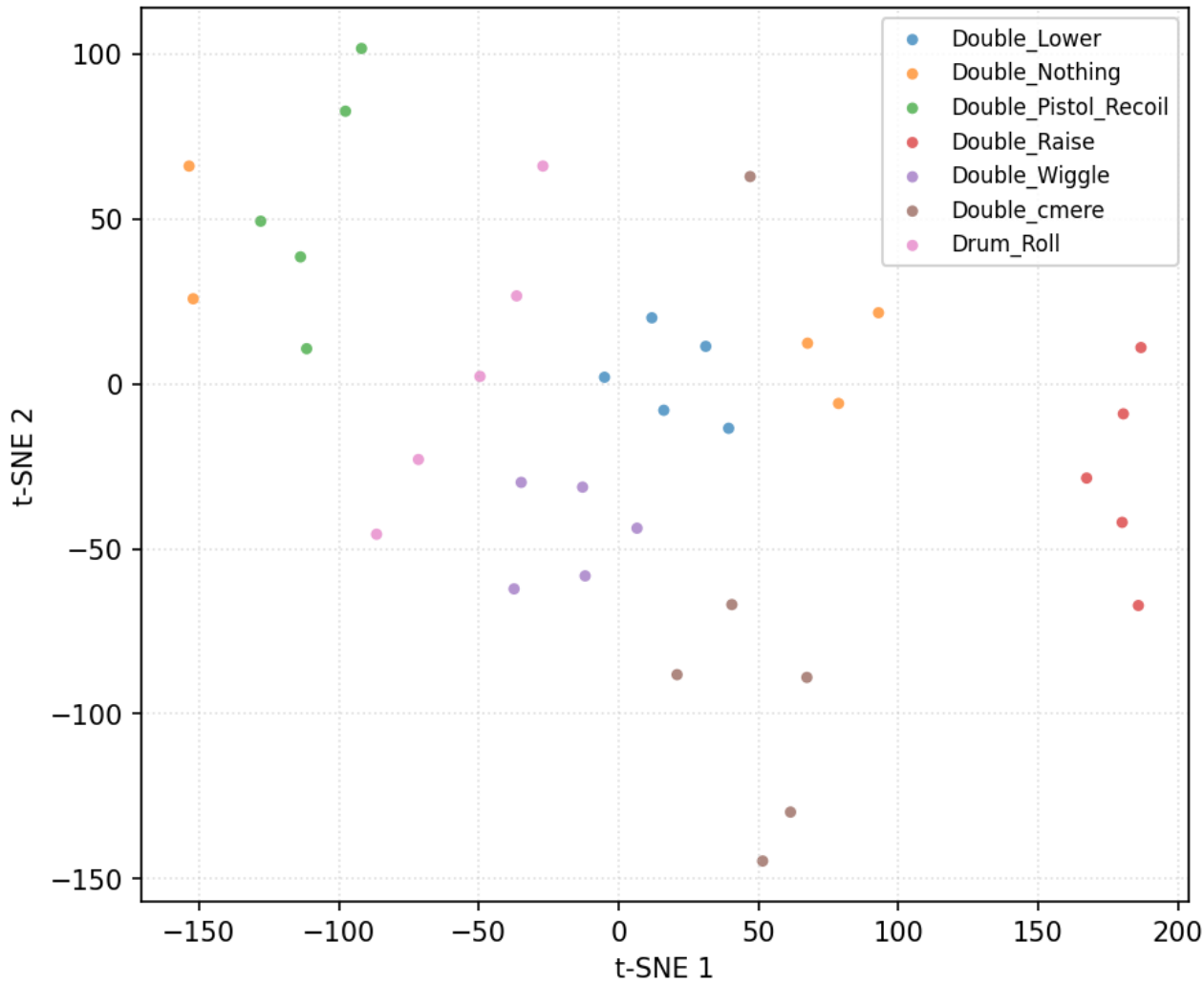


Data diagnostics — user 3_Anghad

36 trials, 7 classes (filtered + resampled, before per-trial normalisation).

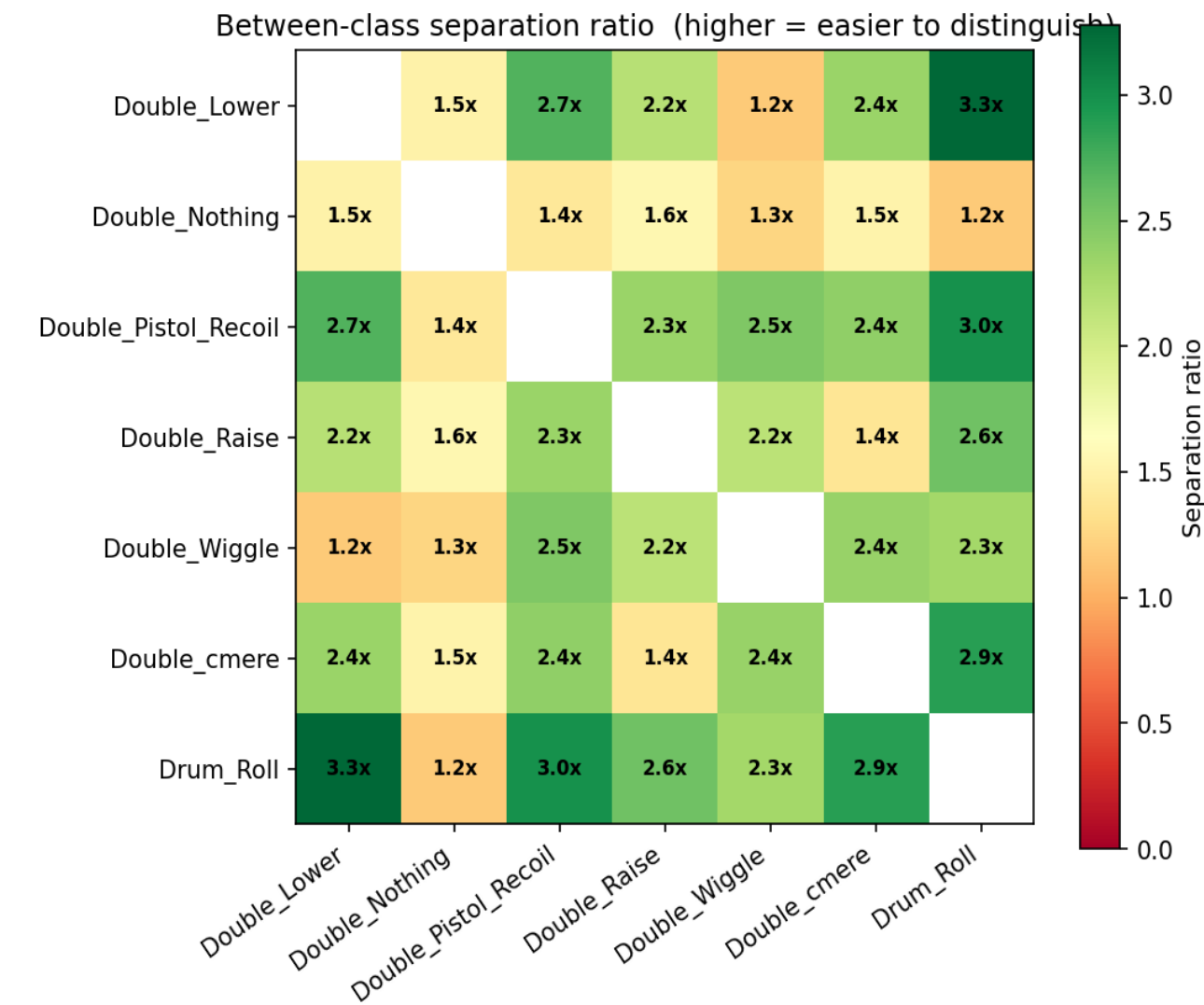
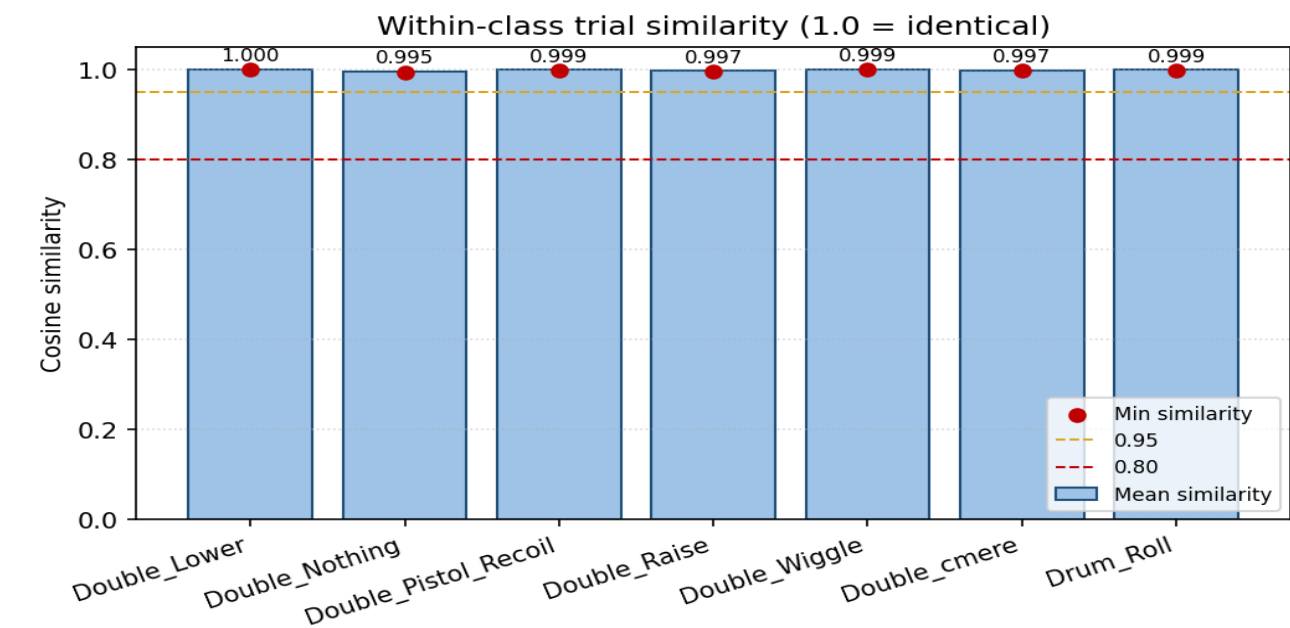


t-SNE feature space projection

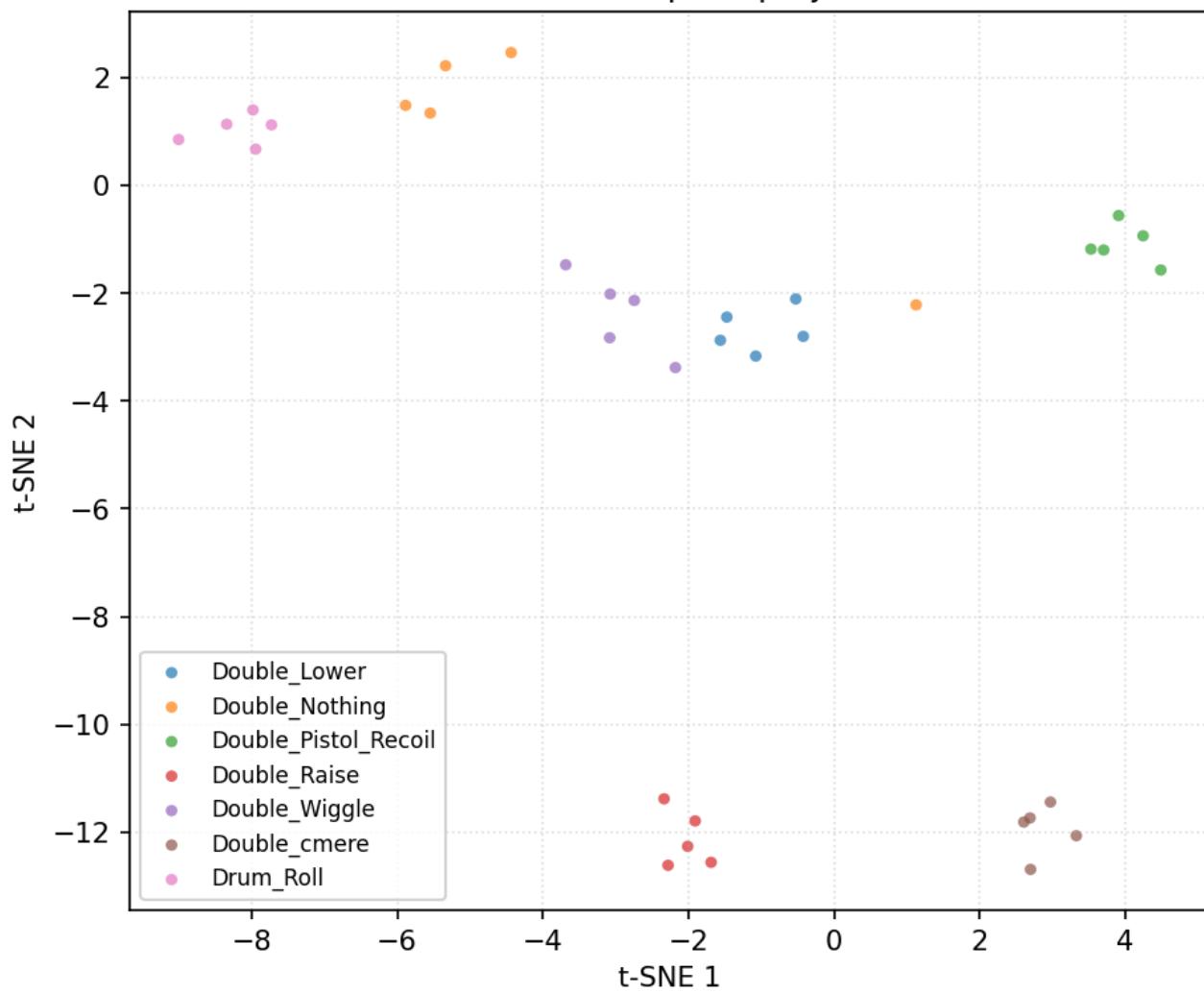


Data diagnostics — user 4_Daniel

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

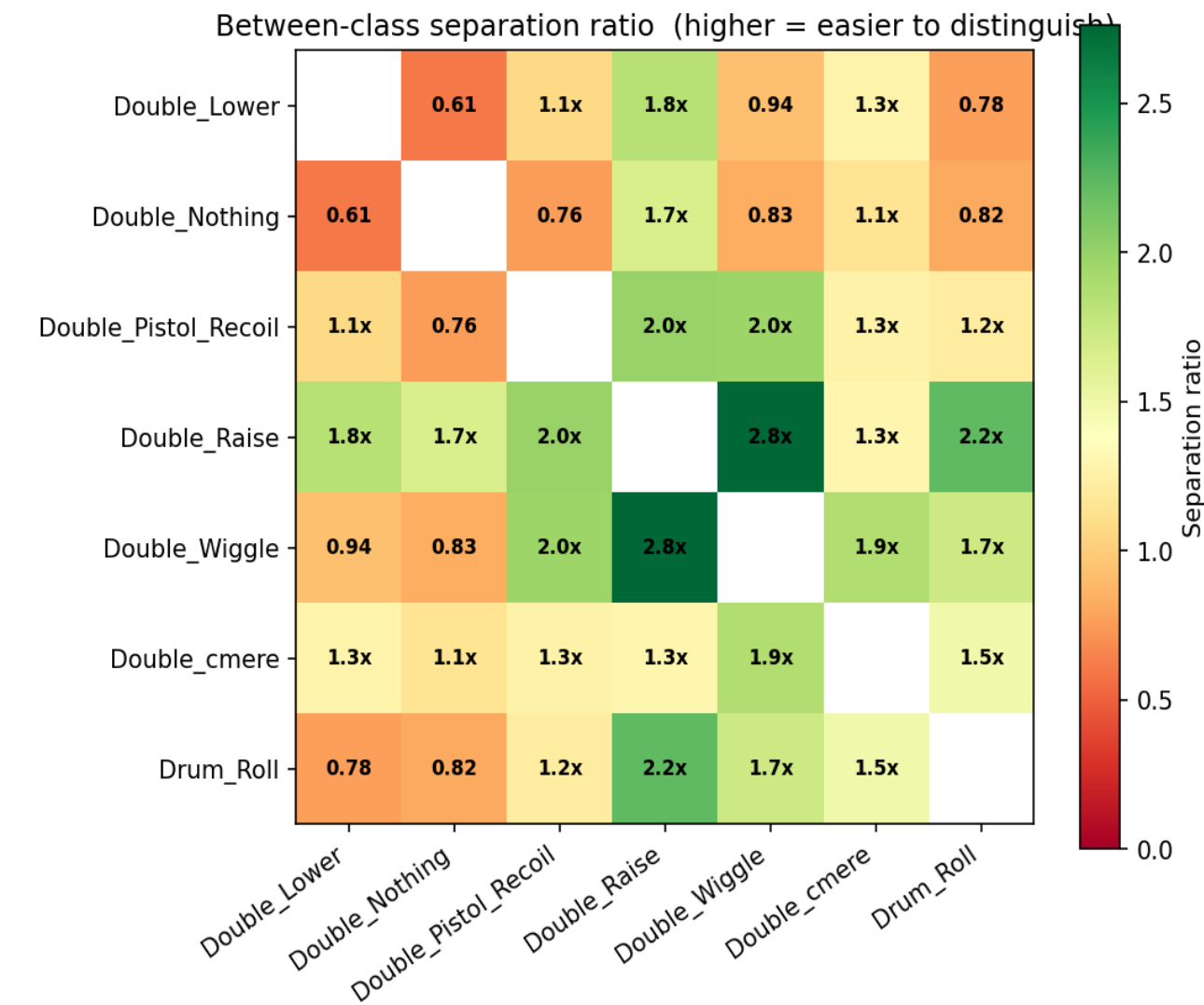
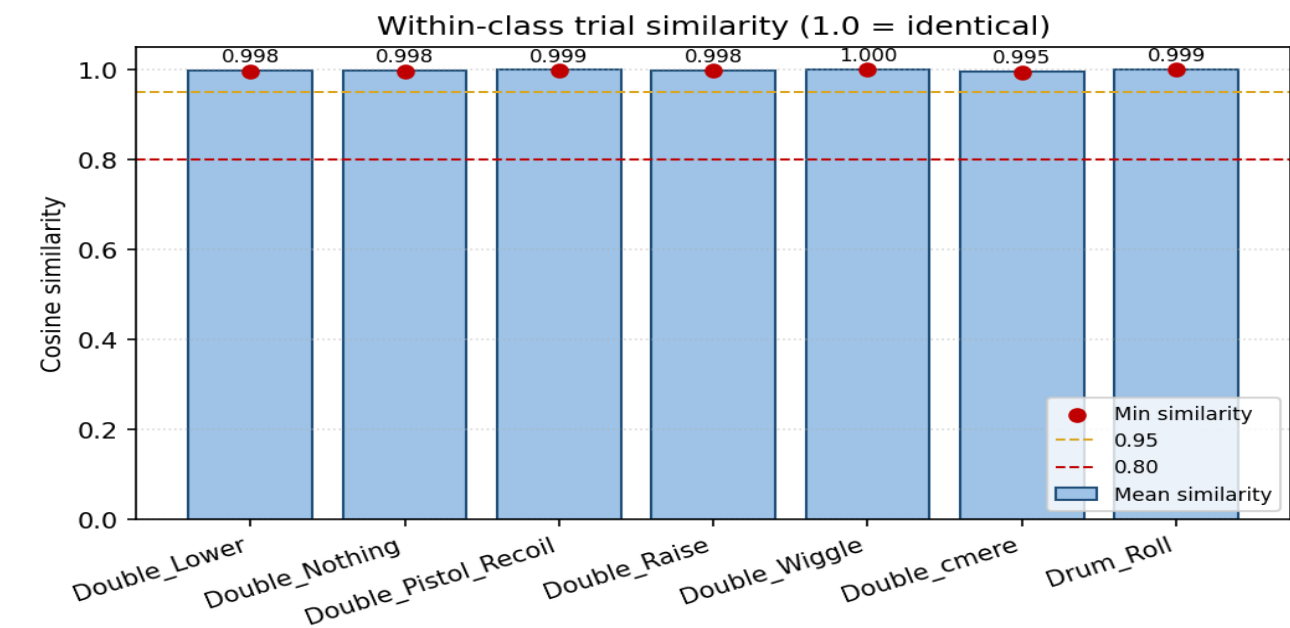


t-SNE feature space projection

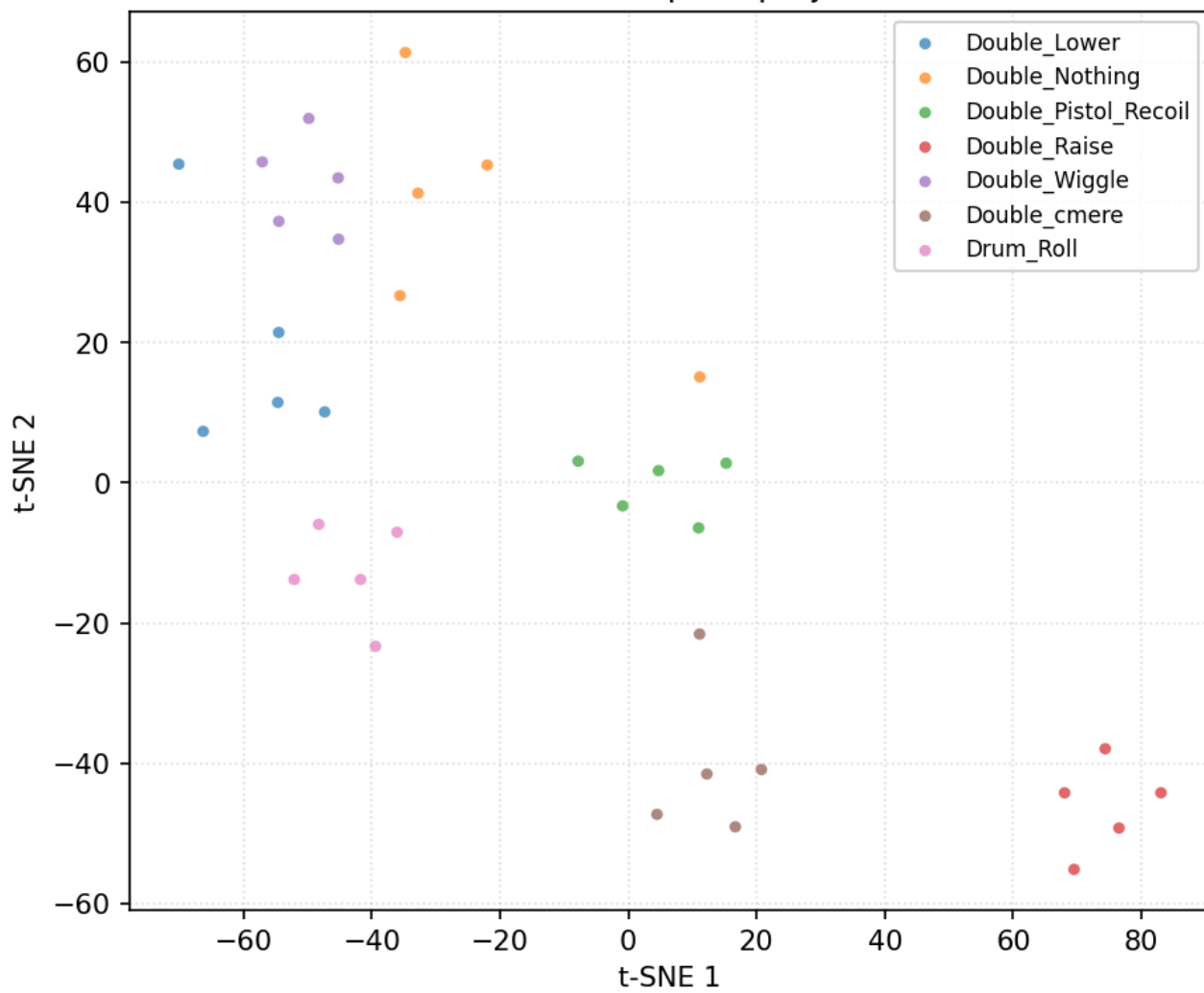


Data diagnostics — user 6_Harry

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

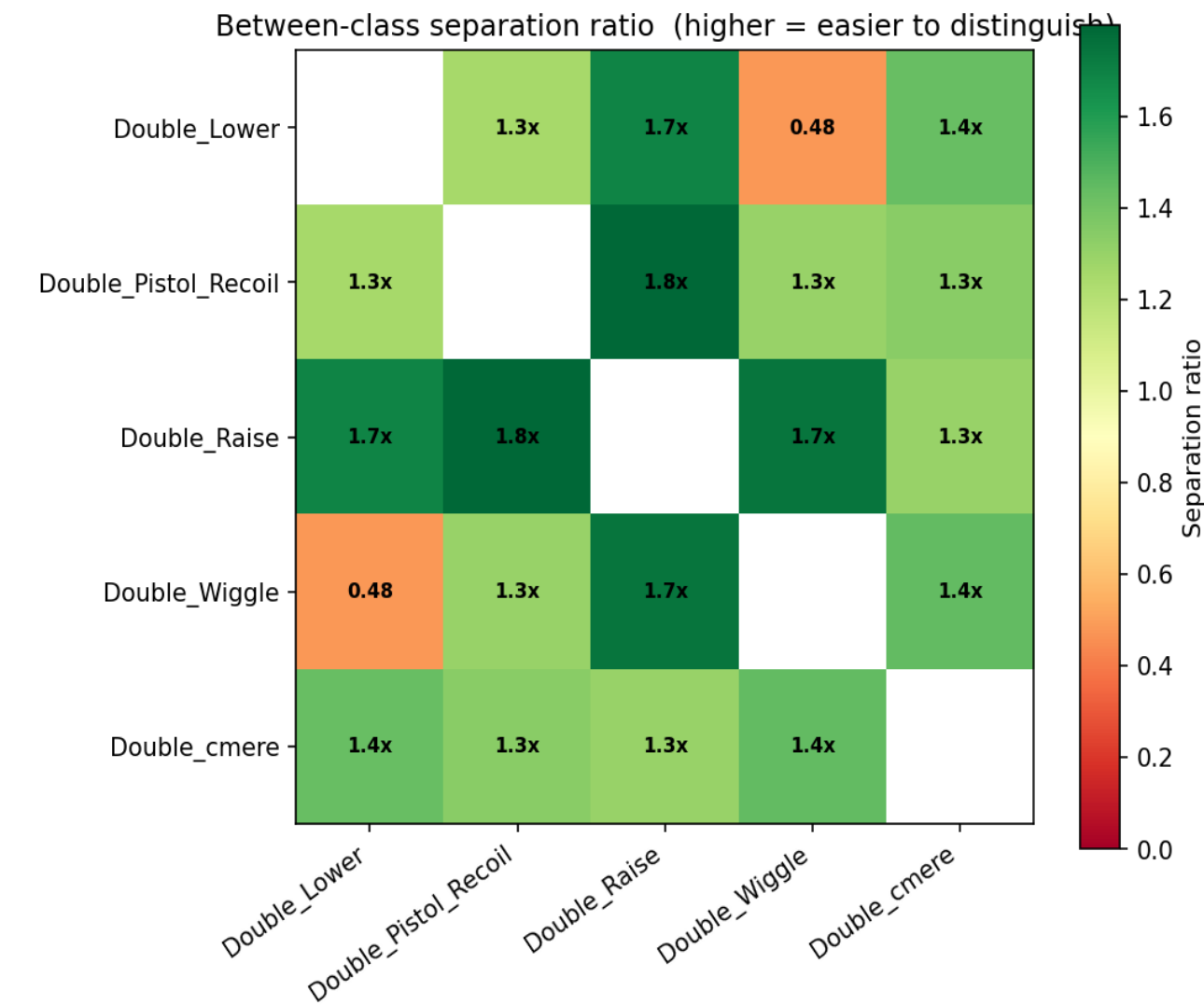
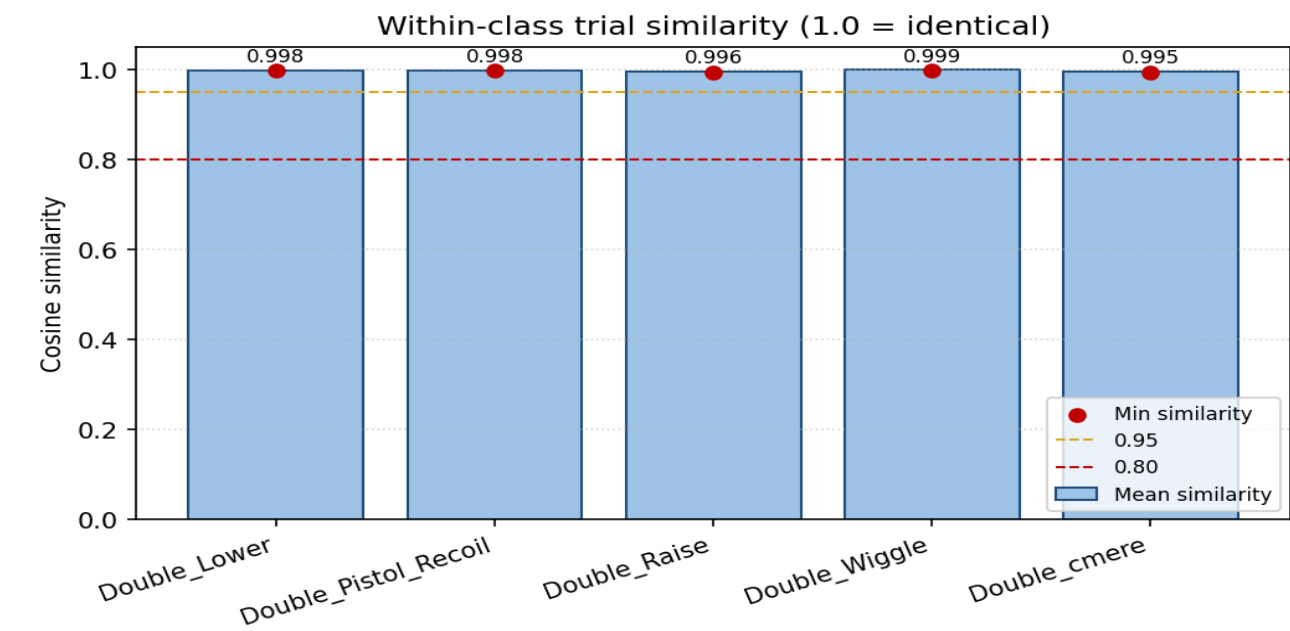


t-SNE feature space projection

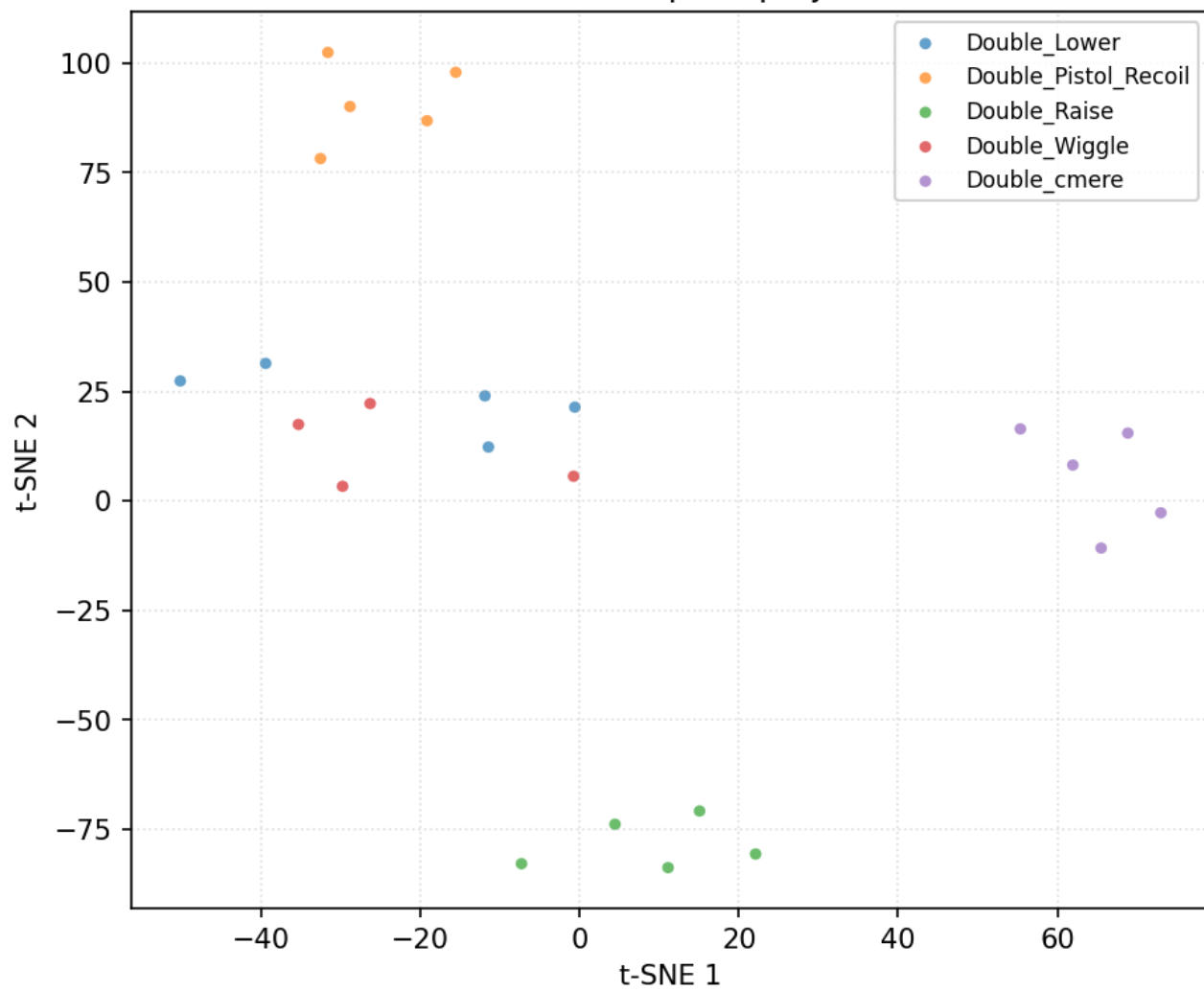


Data diagnostics — user 8_Jestin

24 trials, 5 classes (filtered + resampled, before per-trial normalisation).



t-SNE feature space projection



Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,319
LOUO mean ± std	0.7790 ± 0.1599
Folds run	13
Inner-val mode	user
Per-trial norm	none

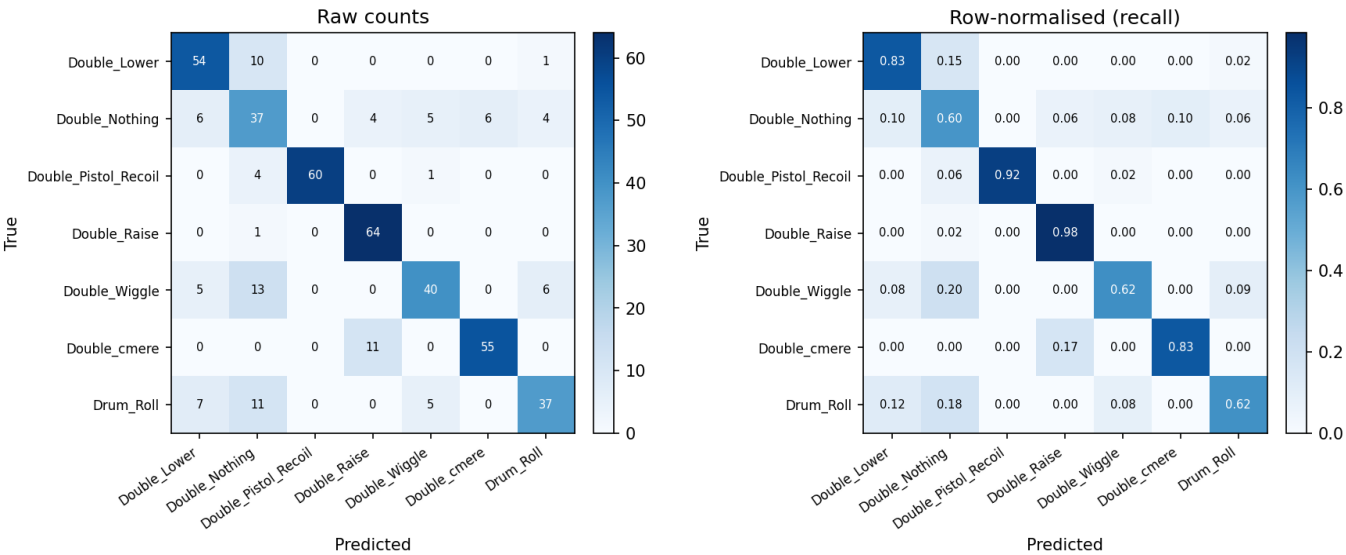
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	377	35	0.8571
2	12_Marcus	14_StephenV2	376	35	0.7429
3	14_StephenV2	15_Tash	376	36	0.8333
4	15_Tash	16_Bella	376	35	0.5714
5	16_Bella	17_Raquel	376	36	0.8611
6	17_Raquel	18_Bridgette	377	35	0.6286
7	18_Bridgette	1_Alan	377	35	0.5429
8	1_Alan	2_Alex	377	35	0.4857
9	2_Alex	3_Anghad	376	35	0.8571
10	3_Anghad	4_Daniel	376	36	0.9444
11	4_Daniel	6_Harry	377	35	0.9714
12	6_Harry	8_Jestin	388	35	0.9143
13	8_Jestin	11_Mansh	388	24	0.9167

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.750	0.831	0.788	65
Double_Nothing	0.487	0.597	0.536	62
Double_Pistol_Recoil	1.000	0.923	0.960	65
Double_Raise	0.810	0.985	0.889	65
Double_Wiggle	0.784	0.625	0.696	64
Double_cmere	0.902	0.833	0.866	66
Drum_Roll	0.771	0.617	0.685	60
macro avg	0.786	0.773	0.774	447
weighted avg	0.789	0.776	0.777	447

Confusion matrix (aggregated across folds)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,255
LOUO mean ± std	0.6350 ± 0.2243
Folds run	13
Inner-val mode	user
Per-trial norm	none

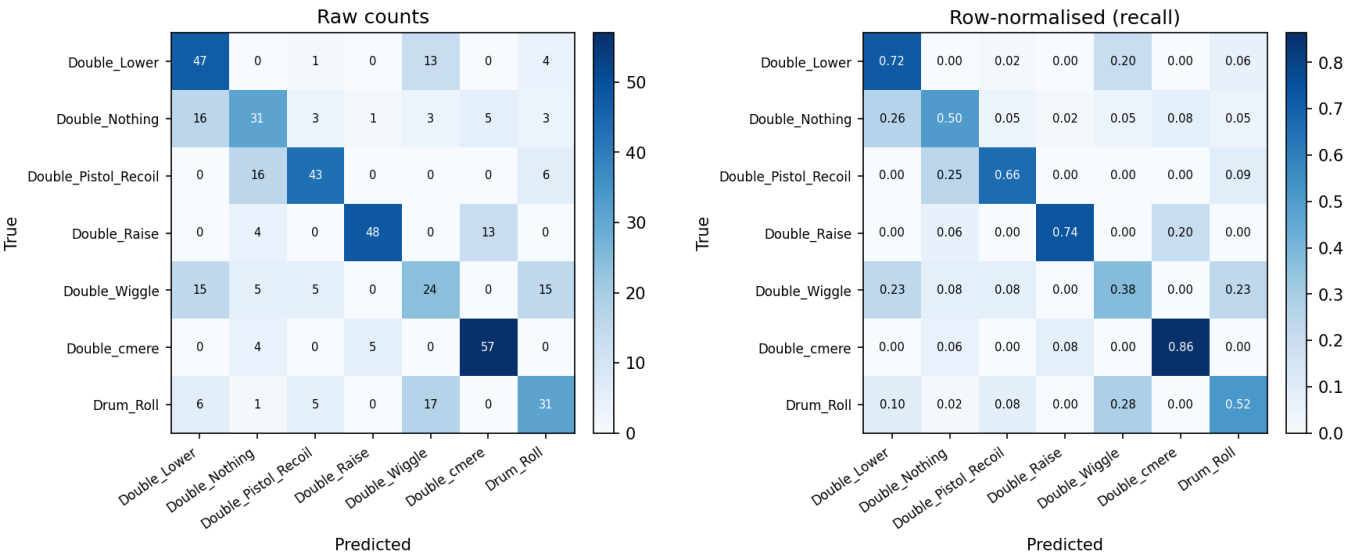
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	377	35	0.6857
2	12_Marcus	14_StephenV2	376	35	0.4286
3	14_StephenV2	15_Tash	376	36	0.7500
4	15_Tash	16_Bella	376	35	0.5429
5	16_Bella	17_Raquel	376	36	0.6389
6	17_Raquel	18_Bridgette	377	35	0.6286
7	18_Bridgette	1_Alan	377	35	0.2857
8	1_Alan	2_Alex	377	35	0.2857
9	2_Alex	3_Anghad	376	35	0.9429
10	3_Anghad	4_Daniel	376	36	0.7500
11	4_Daniel	6_Harry	377	35	0.9714
12	6_Harry	8_Jestin	388	35	0.4286
13	8_Jestin	11_Mansh	388	24	0.9167

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.560	0.723	0.631	65
Double_Nothing	0.508	0.500	0.504	62
Double_Pistol_Recoil	0.754	0.662	0.705	65
Double_Raise	0.889	0.738	0.807	65
Double_Wiggle	0.421	0.375	0.397	64
Double_cmere	0.760	0.864	0.809	66
Drum_Roll	0.525	0.517	0.521	60
macro avg	0.631	0.625	0.625	447
weighted avg	0.634	0.629	0.628	447

Confusion matrix (aggregated across folds)



Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	204,007
LOUO mean ± std	0.6325 ± 0.2225
Folds run	13
Inner-val mode	user
Per-trial norm	none

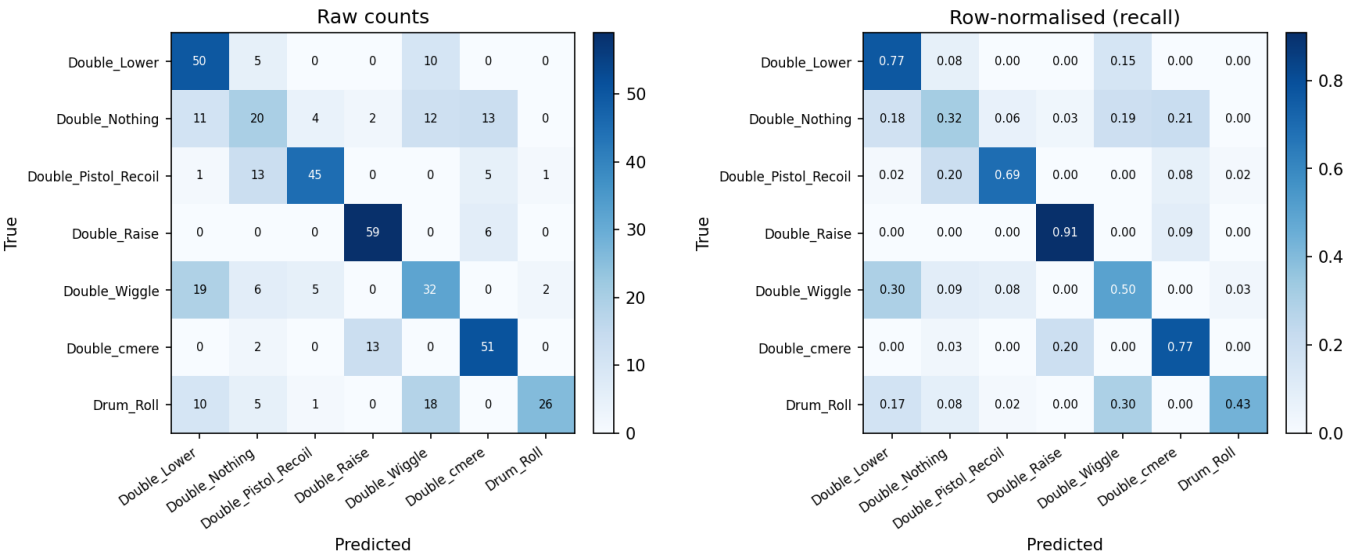
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	377	35	0.6571
2	12_Marcus	14_StephenV2	376	35	0.6857
3	14_StephenV2	15_Tash	376	36	0.8333
4	15_Tash	16_Bella	376	35	0.3143
5	16_Bella	17_Raquel	376	36	0.8333
6	17_Raquel	18_Bridgette	377	35	0.4286
7	18_Bridgette	1_Alan	377	35	0.2857
8	1_Alan	2_Alex	377	35	0.3143
9	2_Alex	3_Anghad	376	35	0.8000
10	3_Anghad	4_Daniel	376	36	0.8889
11	4_Daniel	6_Harry	377	35	0.9429
12	6_Harry	8_Jestin	388	35	0.5714
13	8_Jestin	11_Mansh	388	24	0.6667

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.549	0.769	0.641	65
Double_Nothing	0.392	0.323	0.354	62
Double_Pistol_Recoil	0.818	0.692	0.750	65
Double_Raise	0.797	0.908	0.849	65
Double_Wiggle	0.444	0.500	0.471	64
Double_cmere	0.680	0.773	0.723	66
Drum_Roll	0.897	0.433	0.584	60
macro avg	0.654	0.628	0.625	447
weighted avg	0.654	0.633	0.627	447

Confusion matrix (aggregated across folds)



Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,687
LOUO mean ± std	0.6071 ± 0.1891
Folds run	13
Inner-val mode	user
Per-trial norm	none

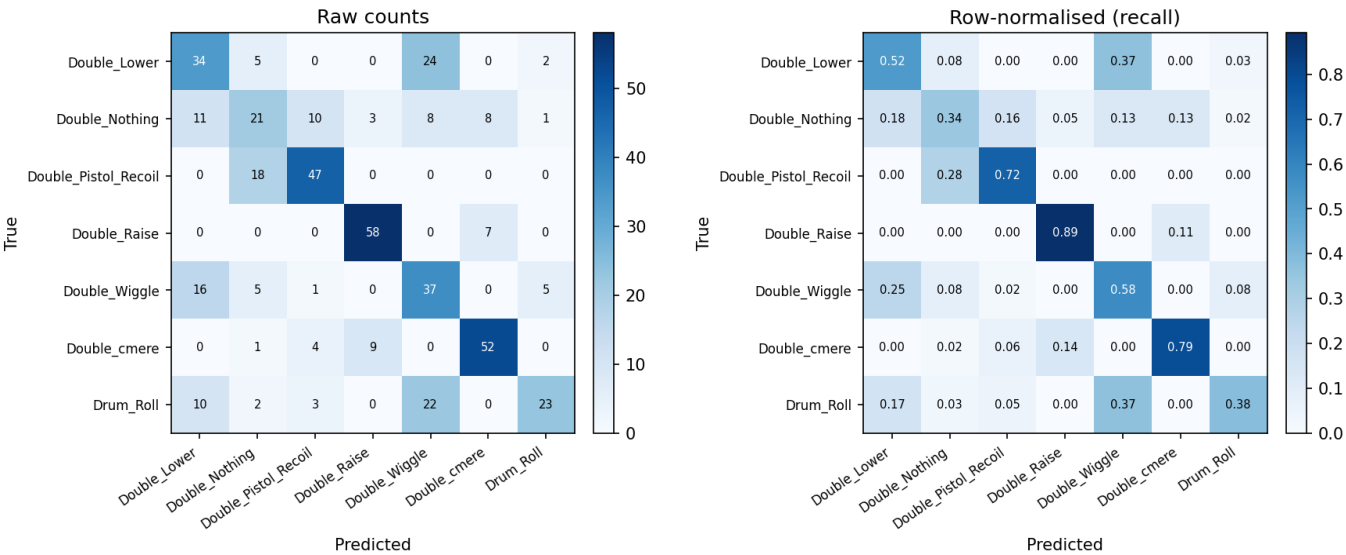
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	377	35	0.5143
2	12_Marcus	14_StephenV2	376	35	0.5143
3	14_StephenV2	15_Tash	376	36	0.6389
4	15_Tash	16_Bella	376	35	0.6286
5	16_Bella	17_Raquel	376	36	0.6111
6	17_Raquel	18_Bridgette	377	35	0.6000
7	18_Bridgette	1_Alan	377	35	0.3143
8	1_Alan	2_Alex	377	35	0.2571
9	2_Alex	3_Anghad	376	35	0.8571
10	3_Anghad	4_Daniel	376	36	0.9167
11	4_Daniel	6_Harry	377	35	0.8857
12	6_Harry	8_Jestin	388	35	0.5714
13	8_Jestin	11_Mansh	388	24	0.5833

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.479	0.523	0.500	65
Double_Nothing	0.404	0.339	0.368	62
Double_Pistol_Recoil	0.723	0.723	0.723	65
Double_Raise	0.829	0.892	0.859	65
Double_Wiggle	0.407	0.578	0.477	64
Double_cmere	0.776	0.788	0.782	66
Drum_Roll	0.742	0.383	0.505	60
macro avg	0.623	0.604	0.602	447
weighted avg	0.624	0.609	0.606	447

Confusion matrix (aggregated across folds)



Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,487
LOUO mean ± std	0.5945 ± 0.1938
Folds run	13
Inner-val mode	user
Per-trial norm	none

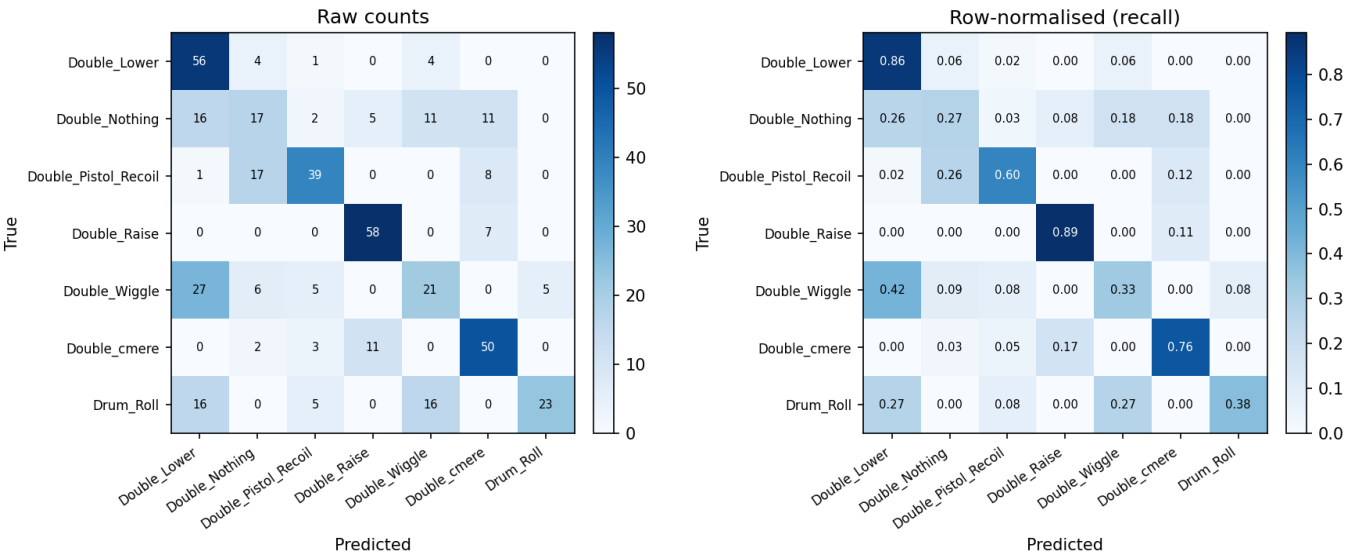
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	377	35	0.6571
2	12_Marcus	14_StephenV2	376	35	0.4857
3	14_StephenV2	15_Tash	376	36	0.8333
4	15_Tash	16_Bella	376	35	0.6000
5	16_Bella	17_Raquel	376	36	0.6389
6	17_Raquel	18_Bridgette	377	35	0.4286
7	18_Bridgette	1_Alan	377	35	0.2000
8	1_Alan	2_Alex	377	35	0.3429
9	2_Alex	3_Anghad	376	35	0.8000
10	3_Anghad	4_Daniel	376	36	0.7222
11	4_Daniel	6_Harry	377	35	0.8000
12	6_Harry	8_Jestin	388	35	0.4286
13	8_Jestin	11_Mansh	388	24	0.7917

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.483	0.862	0.619	65
Double_Nothing	0.370	0.274	0.315	62
Double_Pistol_Recoil	0.709	0.600	0.650	65
Double_Raise	0.784	0.892	0.835	65
Double_Wiggle	0.404	0.328	0.362	64
Double_cmere	0.658	0.758	0.704	66
Drum_Roll	0.821	0.383	0.523	60
macro avg	0.604	0.585	0.572	447
weighted avg	0.604	0.591	0.576	447

Confusion matrix (aggregated across folds)



Variant: CNN_BiLSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→BiLSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	105,575
LOUO mean ± std	0.5921 ± 0.1405
Folds run	13
Inner-val mode	user
Per-trial norm	none

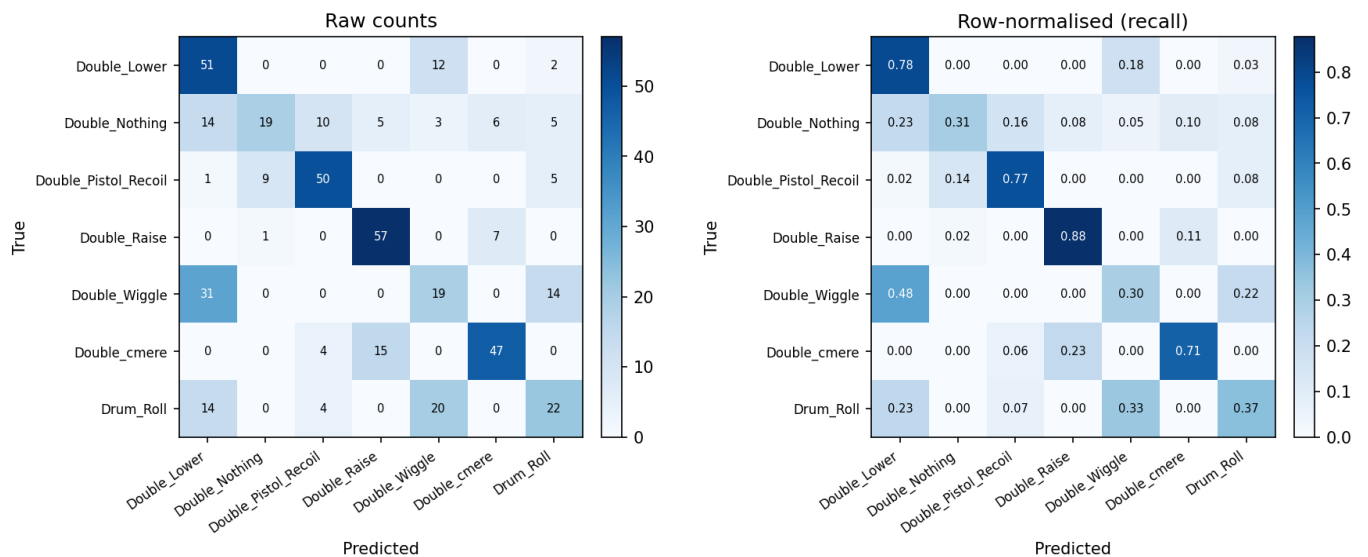
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	377	35	0.5429
2	12_Marcus	14_StephenV2	376	35	0.5143
3	14_StephenV2	15_Tash	376	36	0.5556
4	15_Tash	16_Bella	376	35	0.6857
5	16_Bella	17_Raquel	376	36	0.7222
6	17_Raquel	18_Bridgette	377	35	0.5714
7	18_Bridgette	1_Alan	377	35	0.3714
8	1_Alan	2_Alex	377	35	0.3714
9	2_Alex	3_Anghad	376	35	0.8857
10	3_Anghad	4_Daniel	376	36	0.7222
11	4_Daniel	6_Harry	377	35	0.6857
12	6_Harry	8_Jestin	388	35	0.4857
13	8_Jestin	11_Mansh	388	24	0.5833

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.459	0.785	0.580	65
Double_Nothing	0.655	0.306	0.418	62
Double_Pistol_Recoil	0.735	0.769	0.752	65
Double_Raise	0.740	0.877	0.803	65
Double_Wiggle	0.352	0.297	0.322	64
Double_cmere	0.783	0.712	0.746	66
Drum_Roll	0.458	0.367	0.407	60
macro avg	0.598	0.588	0.575	447
weighted avg	0.600	0.593	0.579	447

Confusion matrix (aggregated across folds)



Variant: CNN_LSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→LSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	68,455
LOUO mean ± std	0.5431 ± 0.2080
Folds run	13
Inner-val mode	user
Per-trial norm	none

LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	377	35	0.5143
2	12_Marcus	14_StephenV2	376	35	0.4857
3	14_StephenV2	15_Tash	376	36	0.6667
4	15_Tash	16_Bella	376	35	0.5429
5	16_Bella	17_Raquel	376	36	0.5556
6	17_Raquel	18_Bridgette	377	35	0.4857
7	18_Bridgette	1_Alan	377	35	0.2857
8	1_Alan	2_Alex	377	35	0.2000
9	2_Alex	3_Anghad	376	35	0.6571
10	3_Anghad	4_Daniel	376	36	0.8889
11	4_Daniel	6_Harry	377	35	0.9429
12	6_Harry	8_Jestin	388	35	0.5429
13	8_Jestin	11_Mansh	388	24	0.2917

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.370	0.569	0.448	65
Double_Nothing	0.467	0.226	0.304	62
Double_Pistol_Recoil	0.637	0.785	0.703	65
Double_Raise	0.797	0.846	0.821	65
Double_Wiggle	0.391	0.391	0.391	64
Double_cmere	0.694	0.652	0.672	66
Drum_Roll	0.500	0.350	0.412	60
macro avg	0.551	0.545	0.536	447
weighted avg	0.553	0.550	0.539	447

Confusion matrix (aggregated across folds)

